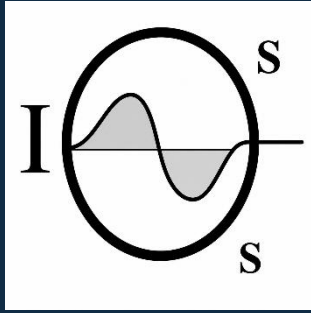




INSPECTION & SPECIFICATION SERVICES CC.

CONTROLLED ENGINEERING INSPECTION REPORT

CABLE BELT MAIN DRIVE SHEAVES

Inspection & Specification Services cc.

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MECHANICAL ASSET CONDITION ASSESSMENT

Cable Belt Main Drive Sheaves Inspection Report

Inspection report · engineering assessment · maintenance recommendations



Client	South32	Mine / site	Wessels Mine
Area / system	Cable Belt Conveyor System	Asset / asset group	Main Drive and Deflector-Guide Sheaves
Report number	IRS32-Wessels-180726	Revision	1.4
Issue date	12/08/2026	Inspection date	18 July 2026
Issue status	Approved	Prepared by	Brendan Wood
Reviewed by	Kevin Wood	Approved by	Kevin Wood
Client reference	South32 - Wessels	ISS job number	IRS32-Wessels-180726

CONFIDENTIALITY & RELIANCE

Prepared for the named client and stated purpose only. Reliance, distribution and operational decisions remain subject to the agreed scope, contractual requirements, stated limitations and responsible engineering review.

Controlled Document Notice

This report records the condition assessment of the South32 Wessels Mine Cable Belt Main Drive and Deflector-Guide Sheaves. It has been compiled from the inspection notes, photographs, recorded field measurements and reference illustrations available at the time of assessment and has been issued in accordance with the document-control and technical-review requirements stated in this report.

The formal alignment/fleet-angle survey and other measurements identified in the recommendations are not prerequisites to issue of this report. They remain open follow-up actions required to quantify the rope-path geometry, define and validate the corrective work, verify the completed intervention and support subsequent finding close-out.

The findings and engineering assessments presented in this report are limited to the agreed inspection scope, access conditions, inspection methods, information supplied and evidence recorded. Visual inspection does not establish the internal condition of bearings, shafts, welds or other inaccessible components unless supported by recorded examination, measurement, condition-monitoring or test evidence.

External standards, OEM requirements, acceptance criteria, measurements and NDT results are relied upon only where they have been verified as applicable to the specific project and are identified accordingly within this report.

Document Control

Table 1 - Document control and report metadata

Report title	Cable Belt Main Drive Sheaves Inspection Report
Report number	IRS32-Wessels-180726
Revision	1.4
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Client	South32
Mine / site	Wessels Mine
Area / system	Cable Belt Conveyor System
Asset / asset group	Main Drive and Deflector-Guide Sheaves
Template version	2.1

Document Authorization Record

Table 2 - Document authorization record

Role	Name	Title / discipline	Email	Date	Initials
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Approved by	Kevin Wood	Managing Engineer	iss@inspecserv.co.za	12/08/2026	KC

Revision History

Table 3 - Revision history

Revision	Date	Issue status	Description of change	Prepared by	Approved by
0.1	02/08/2026	Draft	Preliminary compilation from inspection notes and supplied photographic evidence.	Brendan Wood	Brendan Wood
0.5	08/08/2026	Draft	Engineering review and classification	Brendan Wood	Brendan Wood
1.4	12/08/2026	Approved	Final engineering assessment, system-level alignment recommendations and report issue review completed	Brendan Wood	Kevin Wood

Distribution List

Table 4 - Controlled distribution list

Recipient	Role	Organization	Email	Format	Distribution date
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Alfonso Damon	Planner – Maintenance workforce management planning	South32 – Wessels	Alfonso.Damon@south32.net	PDF	12/08/2026
Kevin Wood	ISS file copy	Inspection & Specification Services cc.	iss@inspecserv.co.za	PDF	12/08/2026

Table of Contents

1 Purpose	9
2 Executive Summary	9
2.1 Executive Dashboard	10
3 Introduction	12
4 Scope of Inspection	12
5 Inspection Limitations	13
6 Applicable Standards and References	14
7 Definitions and Abbreviations	15
8 Responsibilities	16
9 Asset Information	17
9.1 Drive Wheel North Asset Data Sheet	17
9.2 Drive Wheel South Asset Data Sheet	17
9.3 Deflector-Guide Sheave North Asset Data Sheet	18
9.4 Deflector-Guide Sheave South Asset Data Sheet	18
10 Inspection Team and Equipment Used	19
10.1 Inspection Team	19
10.2 Inspection Equipment	19
11 Inspection Methodology	20
12 Inspection Criteria	21
13 Condition Rating, Defect Severity and Maintenance Priority	21
14 Engineering Sketches and Reference Arrangement	23
14.1 Sheave Orientation	23
14.2 Contextual Misalignment Between Drive Wheel Sheave and Deflector-Guide Sheave	24
14.3 Drive Wheel Rope-Entry Fleet Angle	26
14.4 Groove Wear Pattern	27
14.5 Forces on the Sheave Bearings and Shaft	28
14.6 Deflector-Guide Sheave Rope-Exit Fleet Angle	29
15 Asset Descriptions	31
15.1 Drive Wheel North	31
15.2 Drive Wheel South	31
15.3 Deflector-Guide Sheave North	32
15.4 Deflector-Guide Sheave South	32
16 Inspection Findings	33
16.1 Drive Wheel North	33
16.2 Drive Wheel South	40
16.3 Deflector-Guide Sheave North	45
16.4 Deflector-Guide Sheave South	53
17 Summary of Defects	59
18 Inspection Observation Log	60
19 Engineering Assessment	61
20 Recommendations	63

20.1 Corrective Objective	63
20.2 Recommendation Logic	63
20.3 Five-Phase Corrective Sequence	64
20.4 Detailed Recommendation and Action Register	66
20.5 Recommended Sequence - One-Line Summary	67
21 Conclusion	67
22 Technical Review and Report Issue Sign-off	68
Appendix A - Photographic Register	69
Drive Wheel North	72
Drive Wheel South	74
Deflector-Guide Sheave North	76
Deflector-Guide Sheave South	80
Cable Belt Upper Incoming Guide Sheaves	82
Appendix B - Inspection Checklist	83
Appendix C - Measurements and Dimensional Records	85
Appendix D - NDT Results	87
Appendix E - Drawings, Schematics and Reference Information	88
Appendix F - Supporting Documentation	90

List of Figures

Figure 1 - CBDS-REF001 - Cable belt drive sheave arrangement and inspection locations.	23
Figure 2 - CBDS-REF002 - Contextual misalignment mechanism with indicative wear locations.	24
Figure 3 - CBDS-REF003 - Contextual misalignment between the Drive Wheel and Deflector-Guide Sheave.	25
Figure 4 - CBDS-REF004 - Desired alignment with the rope path coincident with the sheave centerlines.	25
Figure 5 - CBDS-REF009 - Desired Drive Wheel Rope Entry Path.	26
Figure 6 - CBDS-REF010 - Contextual Drive Wheel Rope Entry Misalignment.	26
Figure 7 - CBDS-REF008 - Reference groove profile and nominal symmetrical wear pattern illustration.	27
Figure 8 - CBDS-REF005 - Illustrative left-side groove wear produced by angular rope tracking.	27
Figure 9 - CBDS-REF007 - Illustrative bearing-load mechanism associated with a misaligned sheave.	28
Figure 10 - CBDS-REF011 - Desired fleet angle of the rope exiting the Deflector-Guide Sheaves.	29
Figure 11 - CBDS-REF012 - Example of a possible non-zero fleet angle of the rope exiting the Deflector-Guide Sheaves.	29
Figure 12 - CBDS-REF013 - Proposed introduction of lower overhead guide sheaves to manage the exit fleet angle from the Deflector-Guide Sheaves.	30
Figure 13 - Drive Wheel North evidence - PHOTO-DWN-001 / DEF-001.	36
Figure 14 - Drive Wheel North evidence - PHOTO-DWN-002 / DEF-001.	37
Figure 15 - Drive Wheel North evidence - PHOTO-DWN-003 / DEF-001.	38
Figure 16 - Drive Wheel North evidence - PHOTO-DWN-004 / DEF-001.	39
Figure 17 - Drive Wheel South evidence - PHOTO-DWS-001 / DEF-002.	43
Figure 18 - Drive Wheel South evidence - PHOTO-DWS-002 / DEF-002.	44
Figure 19 - Deflector-Guide Sheave North evidence - PHOTO-DGN-002 / DEF-003.	48
Figure 20 - Deflector-Guide Sheave North evidence - PHOTO-DGN-003 / DEF-003.	49
Figure 21 - Deflector-Guide Sheave North evidence - PHOTO-DGN-004 / DEF-003.	50
Figure 22 - Deflector-Guide Sheave North evidence - PHOTO-DGN-005 / DEF-003.	51
Figure 23 - Deflector-Guide Sheave North evidence - PHOTO-DGN-006 / DEF-003.	52
Figure 24 - Deflector-Guide Sheave South evidence - PHOTO-DGS-001 / DEF-004.	56
Figure 25 - Deflector-Guide Sheave South evidence - PHOTO-DGS-002 / DEF-004.	57
Figure 26 - Deflector-Guide Sheave South evidence - PHOTO-DGS-004 / DEF-004.	58
Figure 27 - Drive Wheel North photographic evidence - PHOTO-DWN-001 / PHOTO-DWN-002.	72
Figure 28 - Drive Wheel North photographic evidence - PHOTO-DWN-003 / PHOTO-DWN-004.	73
Figure 29 - Drive Wheel South photographic evidence - PHOTO-DWS-001 / PHOTO-DWS-002.	74
Figure 30 - Drive Wheel South photographic evidence - PHOTO-DWS-003.	75
Figure 31 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-001 / PHOTO-DGN-002.	76
Figure 32 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-003 / PHOTO-DGN-004.	77
Figure 33 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-005 / PHOTO-DGN-006.	78
Figure 34 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-007.	79
Figure 35 - Deflector-Guide Sheave South photographic evidence - PHOTO-DGS-001 / PHOTO-DGS-002.	80
Figure 36 - Deflector-Guide Sheave South photographic evidence - PHOTO-DGS-003 / PHOTO-DGS-004.	81
Figure 37 - Cable Belt Upper Incoming Guide Sheaves contextual arrangement - CBIGS-001.	82
Figure 38 - Cable Belt Upper Incoming Guide Sheaves North / South - CBIGS-002 / CBIGS-003.	82

List of Tables

Table 1 - Document control and report metadata	2
Table 2 - Document authorization record	3
Table 3 - Revision history.....	3
Table 4 - Controlled distribution list.....	3
Table 5 - Executive asset summary matrix	10
Table 6 - Executive recommendation register.....	11
Table 7 - Assets and activities included in scope.....	12
Table 8 - Scope exclusions	12
Table 9 - Inspection limitations register.....	13
Table 10 - Applicable standards and references register.....	14
Table 11 - Definitions and abbreviations	15
Table 12 - Responsibility and interface matrix	16
Table 13 - Drive Wheel North - asset data sheet	17
Table 14 - Drive Wheel South - asset data sheet	17
Table 15 - Deflector-Guide Sheave North - asset data sheet	18
Table 16 - Deflector-Guide Sheave South - asset data sheet.....	18
Table 17 - Inspection team register.....	19
Table 18 - Inspection equipment and calibration register	19
Table 19 - Inspection methodology matrix	20
Table 20 - Inspection criteria and evidence requirements	21
Table 21 - Condition rating.....	21
Table 22 - Defect severity classification	21
Table 23 - Maintenance priority guidance	22
Table 24 - Inspection orientation key	23
Table 25 - Drive Wheel North - asset description summary.....	31
Table 26 - Drive Wheel South - asset description summary	31
Table 27 - Deflector-Guide Sheave North - asset description summary	32
Table 28 - Deflector-Guide Sheave South - asset description summary	32
Table 29 - Drive Wheel North finding classification and action	35
Table 30 - Photographic evidence assessment - PHOTO-DWN-001	36
Table 31 - Photographic evidence assessment - PHOTO-DWN-002	37
Table 32 - Photographic evidence assessment - PHOTO-DWN-003	38
Table 33 - Photographic evidence assessment - PHOTO-DWN-004	39
Table 34 - Drive Wheel South finding classification and action	42
Table 35 - Photographic evidence assessment - PHOTO-DWS-001	43
Table 36 - Photographic evidence assessment - PHOTO-DWS-002	44
Table 37 - Deflector-Guide Sheave North finding classification and action	47
Table 38 - Photographic evidence assessment - PHOTO-DGN-002	48
Table 39 - Photographic evidence assessment - PHOTO-DGN-003	49
Table 40 - Photographic evidence assessment - PHOTO-DGN-004	50
Table 41 - Photographic evidence assessment - PHOTO-DGN-005	51

Table 42 - Photographic evidence assessment - PHOTO-DGN-006	52
Table 43 - Deflector-Guide Sheave South finding classification and action.....	55
Table 44 - Photographic evidence assessment - PHOTO-DGS-001	56
Table 45 - Photographic evidence assessment - PHOTO-DGS-002	57
Table 46 - Photographic evidence assessment - PHOTO-DGS-004	58
Table 47 - Consolidated defect register	59
Table 48 - Inspection observation and evidence log	60
Table 49 - Detailed recommendation and action register	66
Table 50 - Technical review and issue sign-off	68
Table 51 - Photographic register	69
Table 52 - Cable belt sheave inspection checklist	83
Table 53 - Measurements and dimensional records	85
Table 54 - NDT results register	87
Table 55 - Drawings and schematics register	88
Table 56 - Supporting documentation register	90

SECTION A**REPORT BASIS &
GOVERNANCE**

Purpose, executive reporting, scope boundaries, references and responsibility interfaces.



1 Purpose

This report documents the inspection of the Cable Belt Main Drive Sheaves at South32 Wessels Mine. The inspection covered the Drive Wheel North, Drive Wheel South, Deflector-Guide Sheave North and Deflector-Guide Sheave South and was undertaken to record their visible condition, rope-tracking behavior, groove wear patterns and the maintenance actions required to support continued reliable operation.

2 Executive Summary

The inspection evidence is most consistently interpreted as indicating a possible system-level rope-path geometry/alignment condition rather than only four unrelated sheave wear defects. Drive Wheel North shows a visibly off-center rope path together with asymmetric liner deterioration; Drive Wheel South exhibits the opposing asymmetric wear pattern; and both Deflector-Guide Sheaves show laterally displaced, non-symmetrical groove-contact profiles. Taken together, the North and South findings are qualitatively consistent with repeated lateral rope loading through the main drive arrangement. Field notes additionally record as-found Deflector-Guide Sheave groove measurements of 65 mm depth with approximately 3 mm local wear on the North sheave and 66 mm depth with approximately 2 mm local wear on the South sheave. These readings provide quantitative condition context but do not establish remaining service margin or acceptance because verified nominal/new groove geometry and OEM/client discard criteria were not supplied. In addition, the South32 Wessels Mine Condition Monitoring Supervisor reported that vibration analysis had identified an indication consistent with misalignment at Deflector-Guide Sheave North. The underlying vibration trace/report was not supplied for this compilation and is therefore treated as corroborating operational context rather than independently verified measurement evidence.

The consolidated assessment identifies three rope-path interfaces on each side of the system that require quantitative verification: (1) the rope-entry fleet angle from the Cable Belt Upper Incoming Guide Sheaves into Drive Wheel North and South; (2) the fleet angle and centerline relationship between each Drive Wheel and its corresponding Deflector-Guide Sheave; and (3) the downstream fleet angle as the rope exits each Deflector-Guide Sheave. The engineering target is a rope path coincident with the relevant groove centerline, producing a fleet angle of 0 degrees, or as close to 0 degrees as practicable within verified OEM/client tolerances. CBDS-REF009/010 illustrate the Drive Wheel entry condition, CBDS-REF002/003/004 the Drive Wheel-to-Deflector relationship, and CBDS-REF011/012/013 the Deflector-Guide exit condition and conceptual lower overhead guide-sheave option.

Corrective work should therefore be executed as a coordinated sequence: first quantify the complete geometry using a common recorded datum; then correct the rope path by repositioning the Upper Incoming Guide Sheaves and/or Deflector-Guide Sheaves where measured fleet angles are outside the verified tolerance; and, if the downstream Deflector-Guide exit path cannot be brought sufficiently close to zero with the existing arrangement, complete an engineering design review for additional lower overhead guide sheaves. The damaged Drive Wheel liner assemblies on both North and South sides should be renewed, and replacement of both Deflector-Guide Sheaves is recommended because the existing asymmetric groove profiles may no longer provide correct rope support after any confirmed rope-path geometry/alignment error has been corrected. Alignment correction and component renewal are interdependent: replacing worn components without correcting the geometry is likely to reproduce the wear mechanism, while correcting alignment but retaining permanently distorted contact profiles may continue to impose non-uniform support on the 45 mm ropes. The six-interface alignment/fleet-angle survey is the principal follow-up recommendation generated by this report. It remains an open corrective action after report issue and is required to quantify the geometry and define the final repositioning scope; it is not a prerequisite to issue of this inspection report. The four inspected sheaves remain assessed as Poor, Level 3 - Significant, P2 - Urgent. No statement of fitness for continued operation is made; the operational decision remains with the South32 responsible-site authority.

2.1 Executive Dashboard

ASSETS INSPECTED	OVERALL CONDITION	HIGHEST PRIORITY	OPEN ACTIONS
4	Poor	P2 – Urgent	9

Table 5 - Executive asset summary matrix

Asset	Asset tag / ID	Overall condition	Highest severity	Priority	Open findings	Executive comment
Drive Wheel North	North	Poor	Level 3	P2	1	Off-center rope departure and pronounced asymmetric liner damage; verify both Drive Wheel entry fleet angle and North inter-sheave geometry.
Drive Wheel South	South	Poor	Level 3	P2	1	Mirror-image asymmetric liner damage; verify Drive Wheel entry fleet angle and South inter-sheave geometry.
Deflector-Guide Sheave North	North	Poor	Level 3	P2	1	Pronounced off-center groove wear/profile displacement; field notes record 65 mm groove depth and approximately 3 mm local wear. Reported condition-monitoring indication also supports an alignment concern, subject to data review.
Deflector-Guide Sheave South	South	Poor	Level 3	P2	1	Pronounced asymmetric/off-center groove profile; field notes record 66 mm groove depth and approximately 2 mm local wear. Verify both Drive Wheel-to-Deflector geometry and downstream exit fleet angle.

Table 6 - Executive recommendation register

Action ID	Asset	Recommended action	Priority	Responsible discipline	Target / shutdown	Status
REC-001	Complete drive system	Principal follow-up recommendation: perform a quantitative rope-path geometry/alignment survey covering all six entry, inter-sheave and exit fleet-angle interfaces using a common recorded datum.	P2	Mechanical / alignment	Earliest practicable maintenance opportunity	Open
REC-002 to REC-004	Rope-path geometry	Correct measured fleet angles that are outside the verified tolerance by repositioning the Upper Incoming Guide Sheaves and Deflector-Guide Sheaves as required; complete a design review for lower overhead guide sheaves if downstream exit control remains inadequate.	P2	Mechanical / alignment / engineering	Corrective shutdown following survey	Open
REC-005 / REC-006	Drive Wheels and Deflector-Guide Sheaves	Renew the damaged North and South Drive Wheel liner assemblies and replace both Deflector-Guide Sheaves as part of the coordinated geometry-correction intervention.	P2	Mechanical / OEM review	Coordinated corrective shutdown	Open
REC-007 / REC-008	Ropes / bearings / shafts	Inspect both 45 mm ropes and verify bearing/shaft condition; obtain and correlate available condition-monitoring/vibration evidence, particularly for Deflector-Guide Sheave North.	P2	Rope specialist / condition monitoring / mechanical	During corrective intervention	Open
REC-009	Complete corrected system	Verify all corrected fleet angles and rope tracking, capture post-correction groove/profile and condition-monitoring baselines, and retain the records for future trending.	P2	Mechanical / alignment / condition monitoring	Before close-out / return to normal baseline	Open

3 Introduction

Inspection & Specification Services cc. was requested to inspect and document the condition of the cable belt main drive sheaves at South32 Wessels Mine. The inspected arrangement comprises two Drive Wheels and two Deflector-Guide Sheaves, identified as North and South pairs. Each pair guides a nominal 45 mm cable belt rope through the drive arrangement.

The inspection was completed on 18 July 2026. The report provides an evidence-based record of the observed rope path, groove/liner condition and visible wear mechanisms for engineering review, maintenance planning and follow-up action tracking.

4 Scope of Inspection

Table 7 - Assets and activities included in scope

Asset	Asset tag / ID	Location	Inspection activity	Access / operating state	Comment
Drive Wheel North	North	Cable belt main drive station	Visual inspection, photographic documentation, groove/profile assessment and rope-tracking observation	Inspection access state not recorded in supplied notes	Condition assessment for continued operational use.
Drive Wheel South	South	Cable belt main drive station	Visual inspection, photographic documentation, groove/profile assessment and rope-tracking observation	Inspection access state not recorded in supplied notes	Condition assessment for continued operational use.
Deflector-Guide Sheave North	North	Cable belt main drive station	Visual inspection, photographic documentation, groove/profile assessment and rope-tracking observation	Inspection access state not recorded in supplied notes	Condition assessment for continued operational use.
Deflector-Guide Sheave South	South	Cable belt main drive station	Visual inspection, photographic documentation, groove/profile assessment and rope-tracking observation	Inspection access state not recorded in supplied notes	Condition assessment for continued operational use.

Table 8 - Scope exclusions

Excluded item	Reason / access constraint	Assessment implication	Recommended follow-up
Formal geometric alignment survey	No laser/optical alignment readings were supplied.	The presence, magnitude and direction of any geometric misalignment cannot be quantified from photographs alone.	Perform and record a formal alignment survey.
Internal bearing inspection	No bearing disassembly or internal inspection evidence was supplied.	Internal bearing condition cannot be confirmed.	Review condition-monitoring data and inspect bearings where indicated.
NDT	No NDT records were supplied for this report section.	Crack condition of inaccessible/sub-surface regions is not established.	Perform NDT only where required by engineering review / approved scope.
OEM acceptance limits	No verified groove/liner wear limits or alignment tolerances were supplied.	Remaining service margin and acceptance/discard status cannot be established from this evidence alone.	Obtain applicable OEM/client criteria to support dimensional assessment, renewal specification and corrective-action close-out.

5 Inspection Limitations

The assessment is limited to the inspection notes, photographs, recorded field measurements and explanatory reference diagrams supplied for compilation. The photographs provide strong qualitative evidence of asymmetric rope tracking and groove wear, while the inspection field notes provide spot groove-depth and local-wear measurements for the Deflector-Guide Sheaves. The available evidence does not establish the numerical rope-path alignment error, complete circumferential groove geometry, remaining groove/liner service margin, bearing loads or acceptance against a verified OEM/client limit.

Table 9 - Inspection limitations register

Limitation ID	Affected asset / area	Limitation	Impact on assessment	Follow-up requirement
LIM-001	North and South sheave pairs	No formal alignment readings or established datum were supplied.	The observed rope-path and wear patterns are qualitatively consistent with a possible misalignment/fleet-angle mechanism, but the presence, magnitude and direction of any geometric misalignment are not established by the available evidence.	Survey sheave centerlines, shaft axes and rope path using a recorded datum.
LIM-002	Drive Wheel grooves/liners	No dimensional groove-wear measurements or verified liner discard limits were supplied.	Remaining service margin cannot be calculated.	Measure groove profile/liner condition and compare to verified OEM/client limits.
LIM-003	Deflector-Guide Sheaves	Recorded field measurements are available for Deflector-Guide Sheave groove depth and approximate local wear. The Bosch PLR 50, equipment ID 009741746, is recorded as the instrument used for the groove-depth measurements; however, the measurement method/instrument used for the approximate local-wear values was not recorded. The sheave diameter, nominal/new groove geometry, indexed measurement positions and OEM/client groove criteria were not supplied.	The as-found readings can be retained as condition evidence, but remaining wear allowance and acceptance/discard status cannot be calculated from the available data.	Verify the field readings at repeatable indexed positions, confirm and record the measuring equipment and method used for each reading, obtain applicable OEM/client groove geometry and discard criteria, and compare the results.
LIM-004	Bearings, shafts and Deflector-Guide Sheave North condition monitoring	No vibration traces, temperature, end-float, bearing-clearance or shaft-alignment records were supplied. A South32 condition-monitoring indication consistent with Deflector-Guide Sheave North misalignment was reported verbally/contextually but the supporting data were not included for review.	The reported indication provides supporting operational context but does not quantify geometric misalignment or establish bearing/shaft damage.	Obtain and review the relevant vibration trend/analysis, correlate it with the formal alignment survey, and condition-monitor bearings/shafts before and after correction.

6 Applicable Standards and References

No external standard or OEM acceptance limit has been applied as an acceptance basis in this report unless verified as applicable to the specific project. The following supplied drawings and photographs are used as inspection evidence or explanatory reference material. The CBDS reference illustrations are schematic and are not treated as verified dimensional drawings.

Table 10 - Applicable standards and references register

Ref. ID	Type	Title / description	Document number	Revision / date	Source	Status
CBDS-REF001	ISS reference illustration	Cable belt drive sheave arrangement and inspection locations.	CBDS-REF001	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF002	ISS reference illustration	Contextual misalignment mechanism with indicative wear locations.	CBDS-REF002	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF003	ISS reference illustration	Contextual misalignment between the Drive Wheel and Deflector-Guide Sheave.	CBDS-REF003	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF004	ISS reference illustration	Desired alignment with the rope path coincident with the sheave centerlines.	CBDS-REF004	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF005	ISS reference illustration	Illustrative left-side groove wear produced by angular rope tracking.	CBDS-REF005	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF007	ISS reference illustration	Illustrative bearing-load mechanism associated with a misaligned sheave.	CBDS-REF007	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF008	ISS reference illustration	Reference groove profile and nominal symmetrical wear pattern illustration.	CBDS-REF008	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF009	ISS reference illustration	Desired Drive Wheel Rope Entry Path.	CBDS-REF009	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF010	ISS reference illustration	Contextual Drive Wheel Rope Entry Misalignment.	CBDS-REF010	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF011	ISS reference illustration	Desired fleet angle of the rope exiting the Deflector-Guide Sheaves.	CBDS-REF011	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale
CBDS-REF012	ISS reference illustration	Example of a possible non-zero fleet angle of the rope exiting the Deflector-Guide Sheaves.	CBDS-REF012	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - not to scale

Ref. ID	Type	Title / description	Document number	Revision / date	Source	Status
CBDS-REF013	ISS reference illustration	Proposed introduction of lower overhead guide sheaves to manage the exit fleet angle from the Deflector-Guide Sheaves.	CBDS-REF013	Rev 1.0 / supplied 2026	ISS / supplied report package	Reference only - conceptual / not to scale
REF-PHOTO-01	Photographic record	Drive Wheel North inspection photographs	DWN image set	18/07/2026	ISS	Included
REF-PHOTO-02	Photographic record	Drive Wheel South inspection photographs	DWS image set	18/07/2026	ISS	Included
REF-PHOTO-03	Photographic record	Deflector-Guide Sheave North inspection photographs	DGSN image set	18/07/2026	ISS	Included
REF-PHOTO-04	Photographic record	Deflector-Guide Sheave South inspection photographs	DGSS image set	18/07/2026	ISS	Included
REF-PHOTO-05	Photographic record	Cable Belt Upper Incoming Guide Sheave contextual photographs	CBIGS image set	18/07/2026	ISS	Included - context reference

7 Definitions and Abbreviations

Table 11 - Definitions and abbreviations

Term	Definition
IR	Inspection Report
ISS	Inspection & Specification Services cc.
CBDS	Cable Belt Drive Sheaves
DWN	Drive Wheel North
DWS	Drive Wheel South
DGSN	Deflector-Guide Sheave North
DGSS	Deflector-Guide Sheave South
OEM	Original equipment manufacturer
NDT	Non-destructive testing
P1-P4	Maintenance priority classifications defined in this report
CBIGS	Cable Belt Upper Incoming Guide Sheaves

8 Responsibilities

Table 12 - Responsibility and interface matrix

Party / role	Responsibility	Report interface / deliverable
Client management / responsible site authority	Provide scope confirmation, access, isolation, asset information, operational context and formal site decisions.	Review findings, determine operational response and authorize corrective actions.
ISS inspector / project manager	Perform the agreed inspection, record evidence, compile findings and recommendations within the stated limitations.	Prepare and issue controlled inspection reports.
ISS reviewing engineer	Review technical content, evidence traceability, engineering assessment and recommendations.	Technical review and approval.
Maintenance / alignment specialist	Verify alignment and repair scope, execute approved work and provide completion evidence.	Alignment records, repair records, measurements and close-out evidence.

SECTION B

ASSET & INSPECTION
FRAMEWORK

Asset data, inspection equipment, methodology, assessment criteria and reference orientation.



9 Asset Information

The following asset data sheets contain only information supplied in the compilation notes. Unknown OEM, model, bearing and groove-design details are recorded as Not supplied and should be completed from the asset register or verified from OEM documentation held by South32 Wessels Mine.

9.1 Drive Wheel North Asset Data Sheet

Table 13 - Drive Wheel North - asset data sheet

Asset name	Drive Wheel North	Asset tag / ID	North
Location	Cable Belt Main Drive - North	Service / duty	Cable belt main drive wheel - North
Manufacturer / OEM	Not supplied	Model / serial number	Not supplied
Sheave diameter	3200 mm	Number of grooves	Single Groove
Rope diameter	45 mm nominal	Groove profile / liner data	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	2019
Drawing / OEM reference	CBDS-REF001 (arrangement only)	Inspection date	18 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.2 Drive Wheel South Asset Data Sheet

Table 14 - Drive Wheel South - asset data sheet

Asset name	Drive Wheel South	Asset tag / ID	South
Location	Cable Belt Main Drive - South	Service / duty	Cable belt main drive wheel - South
Manufacturer / OEM	Not supplied	Model / serial number	Not supplied
Sheave diameter	3200 mm	Number of grooves	Single Groove
Rope diameter	45 mm nominal	Groove profile / liner data	Not supplied
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	2019
Drawing / OEM reference	CBDS-REF001 (arrangement only)	Inspection date	18 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.3 Deflector-Guide Sheave North Asset Data Sheet

Table 15 - Deflector-Guide Sheave North - asset data sheet

Asset name	Deflector-Guide Sheave North	Asset tag / ID	North
Location	Cable Belt Main Drive - North	Service / duty	Cable belt deflector / guide sheave - North
Manufacturer / OEM	Not supplied	Model / serial number	Not supplied
Sheave diameter	Not supplied	Number of grooves	Single Groove
Rope diameter	45 mm nominal	Groove profile / liner data	Field record: groove depth 65 mm; approximately 3 mm local groove wear (inspection field notes).
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	2019
Drawing / OEM reference	CBDS-REF001 (arrangement only)	Inspection date	18 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

9.4 Deflector-Guide Sheave South Asset Data Sheet

Table 16 - Deflector-Guide Sheave South - asset data sheet

Asset name	Deflector-Guide Sheave South	Asset tag / ID	South
Location	Cable Belt Main Drive - South	Service / duty	Cable belt deflector / guide sheave - South
Manufacturer / OEM	Not supplied	Model / serial number	Not supplied
Sheave diameter	Not supplied	Number of grooves	Single Groove
Rope diameter	45 mm nominal	Groove profile / liner data	Field record: groove depth 66 mm; approximately 2 mm local groove wear (inspection field notes).
Bearing type	Not supplied	Shaft / axle reference	Not supplied
Design speed / duty	Not supplied	Commissioned / installation date	2019
Drawing / OEM reference	CBDS-REF001 (arrangement only)	Inspection date	18 July 2026
Inspector(s)	ISS inspection team	Overall condition rating	Poor

10 Inspection Team and Equipment Used

10.1 Inspection Team

Table 17 - Inspection team register

Name	Role / discipline	Company	Competency / authorization	Site induction / permit reference
Brendan Wood	Inspector / Projects and Systems	ISS	Technician	N/A
Kevin Wood	Technical reviewer / Managing Engineer	ISS	Managing Engineer	N/A

10.2 Inspection Equipment

Table 18 - Inspection equipment and calibration register

Equipment	Purpose	Manufacturer / model	Equipment ID	Calibration certificate	Calibration due	Accuracy / range	Comment
Digital camera / mobile device	Photographic documentation	Not recorded	Not recorded	N/A	N/A	N/A	Used for visual evidence record.
Laser distance meter	Groove-depth reading	Bosch PLR 50	009741746	Not supplied	Not verified	Manufacturer accuracy/range not verified from supplied documentation	Used in the sheave groove depth measurements.
Steel rule	Contextual indication of rope path offset at Drive Wheel North	Not recorded	Not recorded	Not recorded	Not recorded	Not recorded	Photograph provides contextual scale only; no formal alignment value recorded.
Groove profile	Comparative assessment of Deflector-Guide Sheave groove profile	Not recorded	Not recorded	Not recorded	Not recorded	Profile not supplied	Gauge contact is visible in DGSN/DGSS photographs. Separate inspection field notes record Deflector-Guide groove-depth/local-wear values. The Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurements; the groove-profile gauge ID and the measurement method/instrument used for the approximate local-wear values were not recorded.

11 Inspection Methodology

The inspection methodology comprised visual examination, photographic documentation, comparative rope-groove profile checks and qualitative observation of the rope approach/departure relationship between each Drive Wheel and corresponding Deflector-Guide Sheave. No formal laser/optical alignment survey or NDT record was supplied for this report section.

Table 19 - Inspection methodology matrix

Method ID	Inspection activity	Purpose	Evidence captured	Acceptance / reference basis	Limitation / note
M-01	Visual inspection	Record visible groove, liner and adjacent component condition.	Inspection notes and photographs	Engineering assessment / verified OEM criteria where available	Limited to accessible visible surfaces.
M-02	Rope tracking observation	Record lateral rope position and off-center entry/exit indicators.	DWN/DWS photographs and CBDS reference illustrations	Formal alignment datum required for quantitative acceptance	Photographic evidence is qualitative.
M-03	Groove profile assessment	Compare accessible groove profile using the supplied profile gauge.	DGSN/DGSS profile-gauge photographs and inspection field-note readings: North 65 mm groove depth / approximately 3 mm local wear; South 66 mm groove depth / approximately 2 mm local wear.	Verified OEM groove geometry / discard limits required	The Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurements. The groove-profile gauge ID and the measurement method/instrument used for the approximate local-wear values were not recorded; wear values are approximate and OEM/client acceptance criteria were not supplied.
M-04	Wear mechanism review	Relate observed asymmetric wear to the rope path and sheave relationship.	Photographs and CBDS-REF002 to REF008	Engineering interpretation within stated limitations	Illustrations are explanatory and not dimensional drawings.

12 Inspection Criteria

Table 20 - Inspection criteria and evidence requirements

Attribute	Assessment basis	Evidence required	Typical record	Reference / note
Rope groove / liner condition	Visible sidewall wear, material loss, asymmetric polishing, profile distortion and localized damage.	Photographs and profile-gauge evidence.	Finding / defect ID	Final acceptance requires verified OEM/client criteria.
Rope tracking and alignment	Rope position relative to groove centerline at sheave entry/exit and evidence of angled tracking.	Photographs and formal alignment survey where performed.	Finding / defect ID	Photographs support qualitative assessment only.
Deflector-Guide groove profile	Gauge fit, off-center contact and sidewall wear pattern.	Profile-gauge photographs, inspection field-note measurements and repeatable dimensional records.	Finding / defect ID	Field-note readings are available; the Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurements. The groove-profile gauge ID and measurement method/instrument used for the approximate local-wear values were not recorded; nominal/new groove geometry and verified OEM/client discard criteria were not supplied.
Hub, shaft and bearings	Visible security/leakage indicators and condition-monitoring evidence where available.	Photographs, vibration/temperature/alignment records.	Observation / follow-up	No bearing data supplied.
Structural / brake context	Visible general condition of accessible supports and brake assemblies.	Context photographs / notes.	Observation ID	Not the primary defect mechanism in supplied notes.

13 Condition Rating, Defect Severity and Maintenance Priority

The following classifications support consistent reporting and maintenance planning. They do not replace the mine's formal risk assessment, statutory obligations or decisions by the responsible site authority.

Table 21 - Condition rating

Rating	General meaning	Typical reporting response
Excellent	No material defect observed within the inspection scope.	Record condition and routine inspection interval.
Good	Minor condition observations only.	Monitor through routine maintenance and inspection.
Fair	Serviceable with deterioration requiring planned attention.	Plan maintenance, measurement or enhanced monitoring.
Poor	Significant deterioration requiring priority engineering or maintenance review.	Define corrective action and verification requirements.
Critical	Potentially unacceptable condition requiring immediate responsible-site review.	Escalate immediately and determine operational response.

Table 22 - Defect severity classification

Level	Classification	Definition	Default action
Level 1	Minor	Low-significance observation requiring monitoring or routine correction.	Record and monitor.
Level 2	Moderate	Defect requiring planned corrective action.	Schedule corrective work.
Level 3	Significant	Defect requiring priority engineering review and corrective action.	Repair at the next suitable shutdown or earlier.
Level 4	Critical	Potentially critical condition requiring immediate responsible-site review.	Immediate escalation and corrective action.

Table 23 - Maintenance priority guidance

Priority	Meaning	Typical timeframe guidance	Required response
P1	Immediate	Before continued operation where required by responsible review.	Escalate and define immediate controls / corrective action.
P2	Urgent	Earliest practicable maintenance opportunity.	Plan and execute short-term corrective action.
P3	Planned	Next suitable scheduled shutdown or maintenance cycle.	Include in approved maintenance plan.
P4	Monitor	Routine inspection and maintenance interval.	Trend and verify at future inspection.

14 Engineering Sketches and Reference Arrangement

The following ISS reference illustrations are used to explain the inspected arrangement, contextual misalignment mechanisms, resulting groove wear patterns and possible load transfer to the bearings. They are schematic, not to scale and are not substitutes for a verified as-built alignment drawing or measured alignment survey.

14.1 Sheave Orientation

CBDS-REF001 identifies the North and South Drive Wheels and their corresponding Deflector-Guide Sheaves within the cable belt drive arrangement.

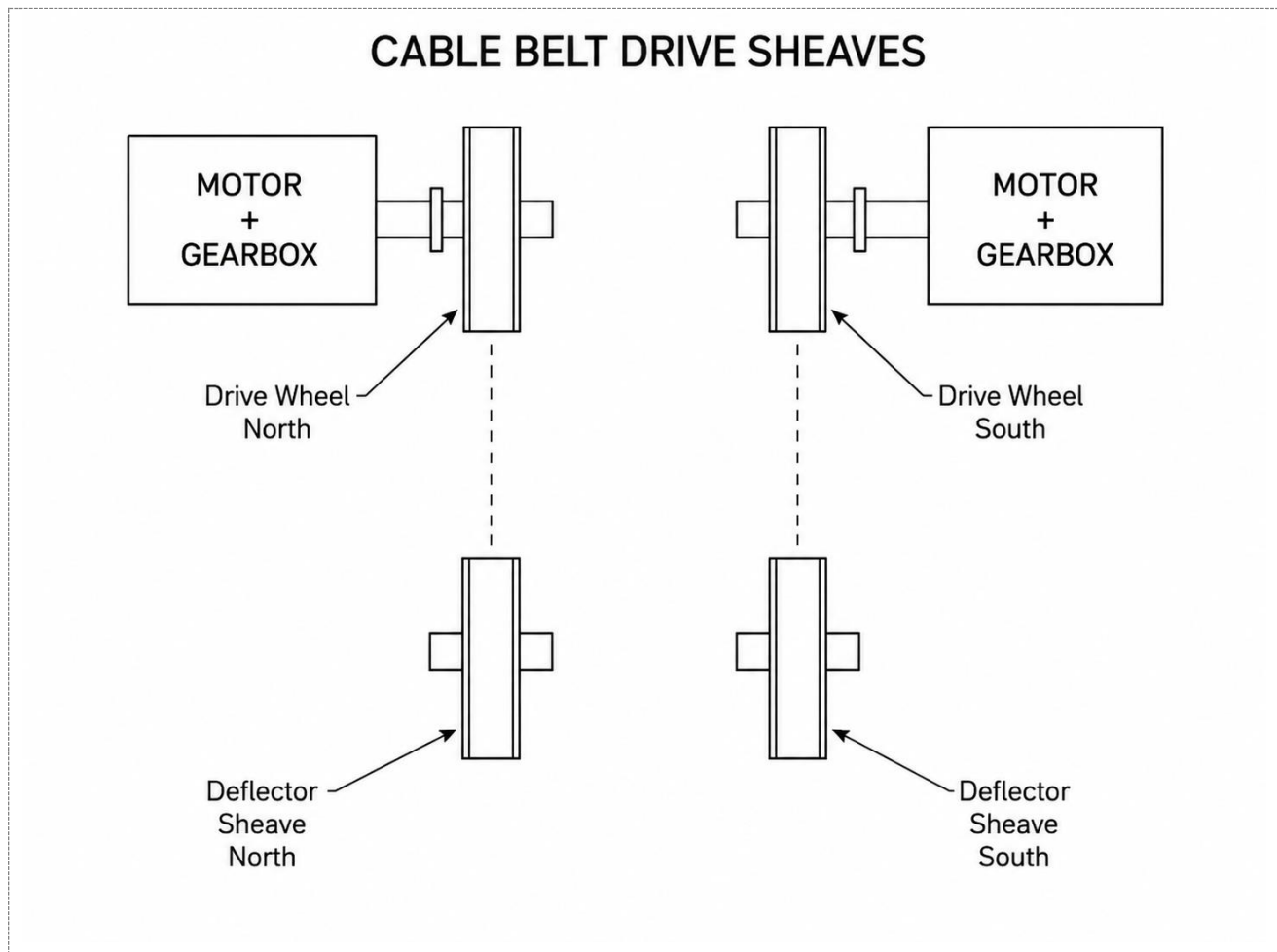


Figure 1 - CBDS-REF001 - Cable belt drive sheave arrangement and inspection locations.

Table 24 - Inspection orientation key

Reference	Description	Direction / datum	Source drawing / photograph
DWN	Drive Wheel North	North drive path	CBDS-REF001
DGSN	Deflector-Guide Sheave North	North guide path / paired with DWN	CBDS-REF001
DWS	Drive Wheel South	South drive path	CBDS-REF001
DGSS	Deflector-Guide Sheave South	South guide path / paired with DWS	CBDS-REF001

14.2 Contextual Misalignment Between Drive Wheel Sheave and Deflector-Guide Sheave

The contextual misalignment illustrations depict a hypothetical condition in which the Deflector-Guide Sheave is offset from the Drive Wheel centerline. In the illustrated condition, the rope path becomes angular, which could result in off-center rope departure from the Drive Wheel and off-center entry into the Deflector-Guide Sheave. CBDS-REF004 illustrates the desired centered relationship.

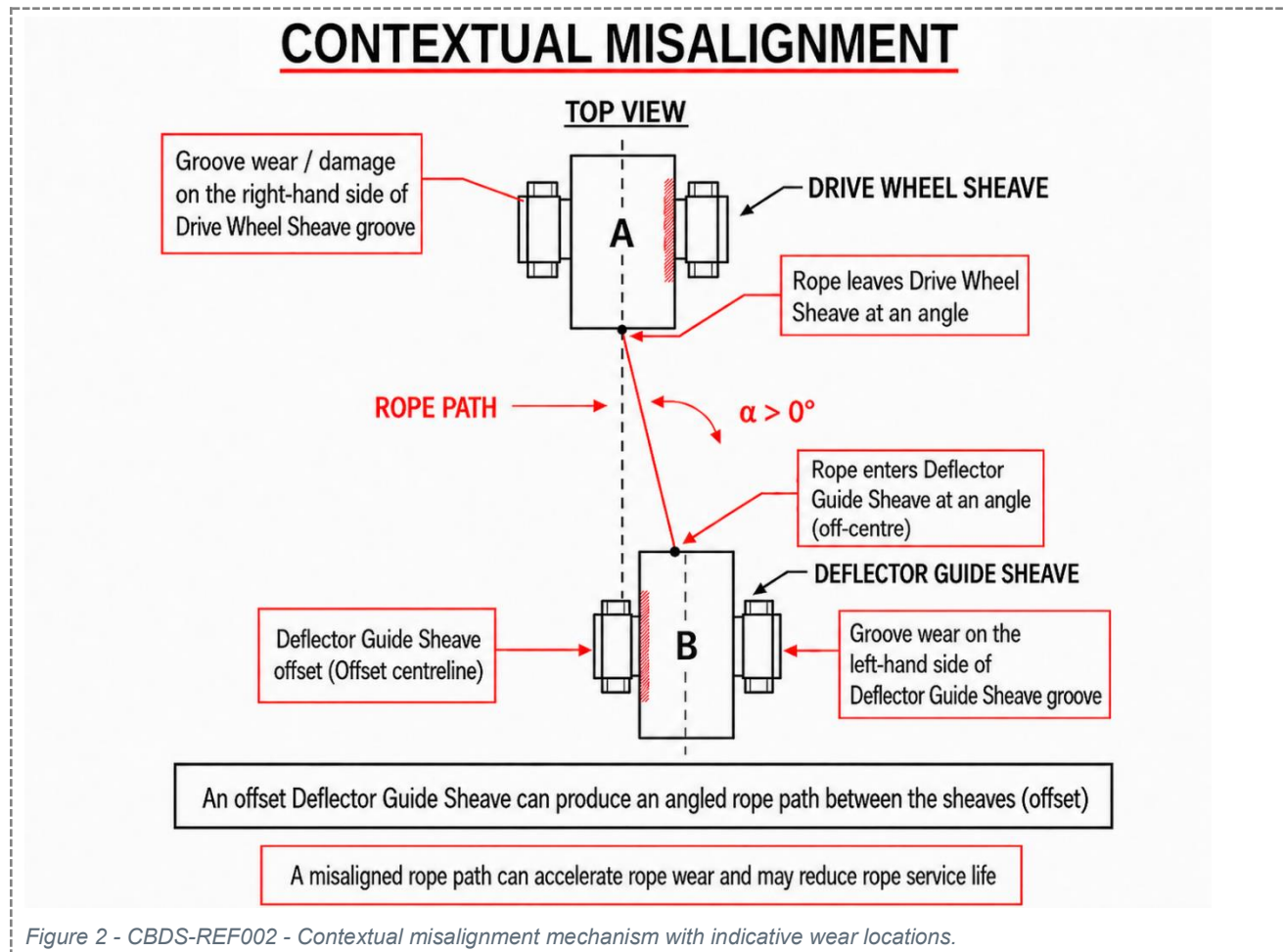


Figure 2 - CBDS-REF002 - Contextual misalignment mechanism with indicative wear locations.

CONTEXTUAL MISS-ALIGNMENT

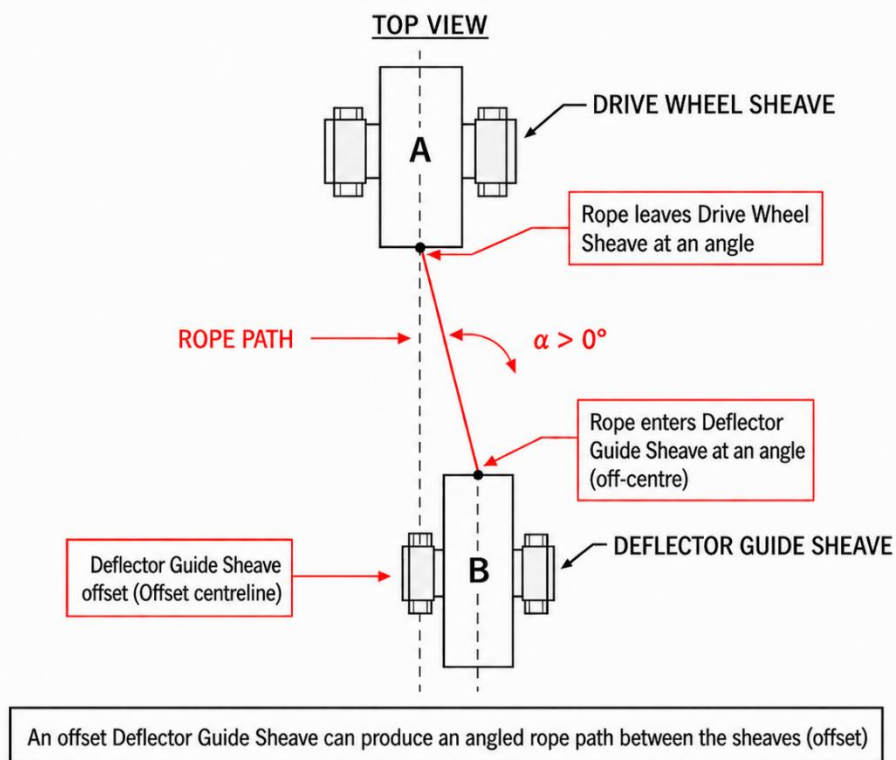


Figure 3 - CBDS-REF003 - Contextual misalignment between the Drive Wheel and Deflector-Guide Sheave.

DESIRED ALIGNMENT

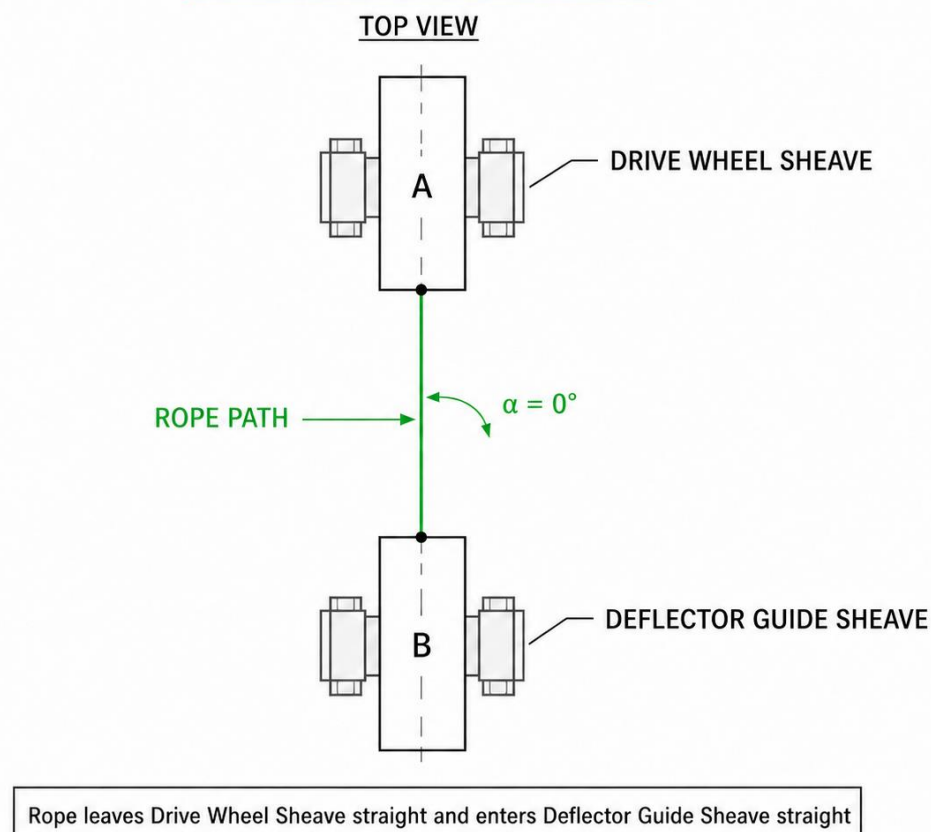


Figure 4 - CBDS-REF004 - Desired alignment with the rope path coincident with the sheave centerlines.

14.3 Drive Wheel Rope-Entry Fleet Angle

The following schematic illustrations show the desired rope-entry condition at the Drive Wheel sheaves, where the fleet angle α is 0° or as close to that condition as practicable, and a contextual example of a possible non-zero rope-entry fleet angle. These figures are explanatory sketches only, are not to scale, and do not constitute a measured alignment or as-built survey. They are included to support the later system-level interpretation and recommendations.

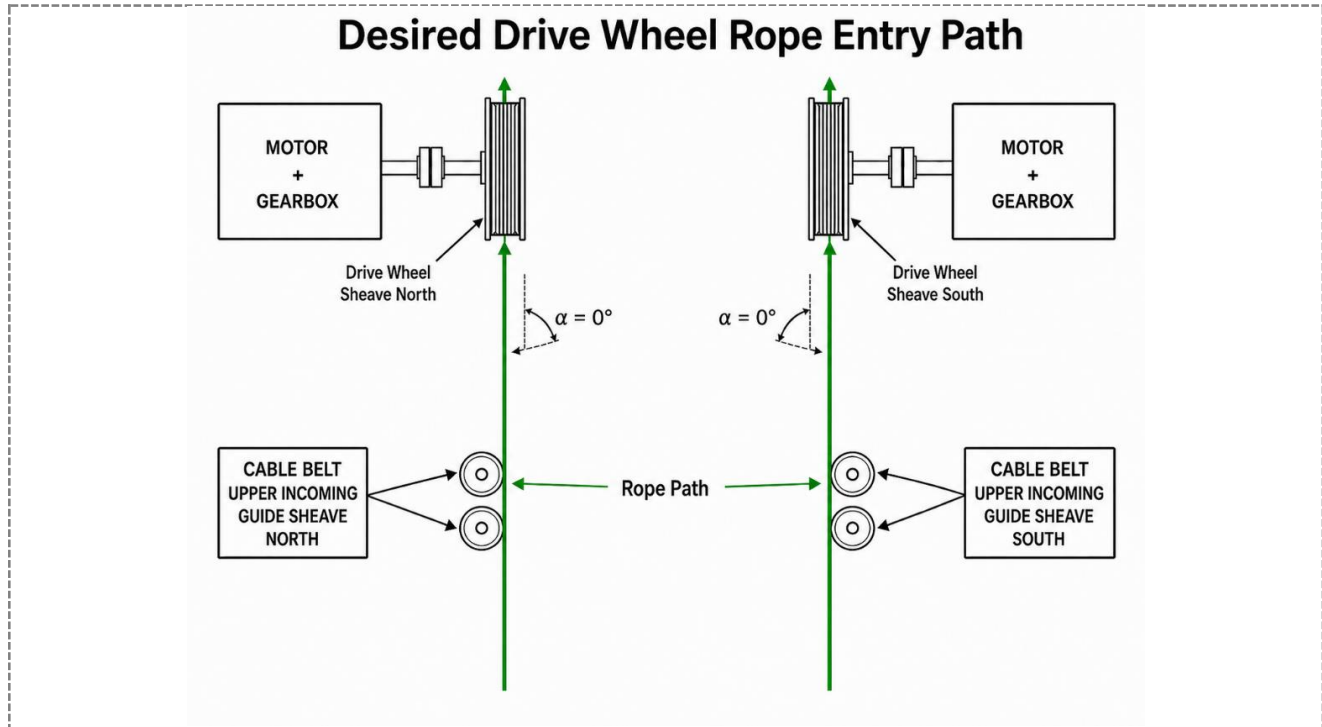


Figure 5 - CBDS-REF009 - Desired Drive Wheel Rope Entry Path.

CONTEXTUAL DRIVE WHEEL ROPE ENTRY MISALIGNMENT

TOP VIEW - ROPE ENTRY PATH

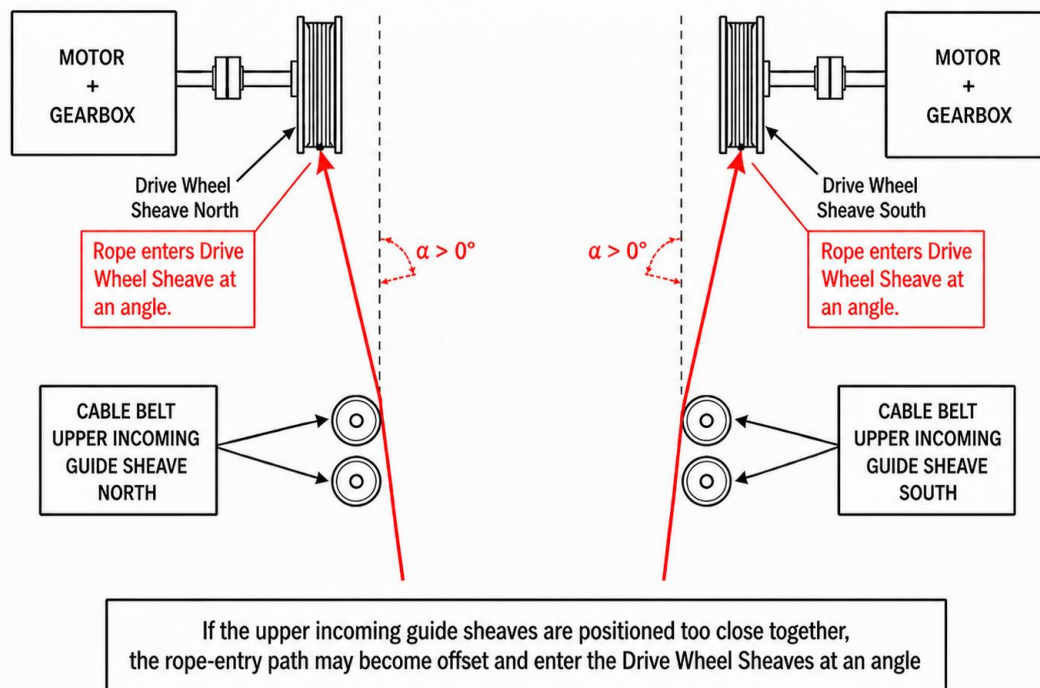


Figure 6 - CBDS-REF010 - Contextual Drive Wheel Rope Entry Misalignment.

14.4 Groove Wear Pattern

The supplied wear-pattern illustrations distinguish a nominal, substantially symmetrical groove contact pattern from localized sidewall wear that may occur under angular or off-center rope tracking. The actual photographs should be used as the primary inspection evidence. These diagrams provide explanatory context only and are not used as acceptance criteria in this report. Any allowable groove-wear limit or D:d-related (Sheave Diameter/rope Diameter) criterion must be verified against the applicable OEM/client requirements before use.

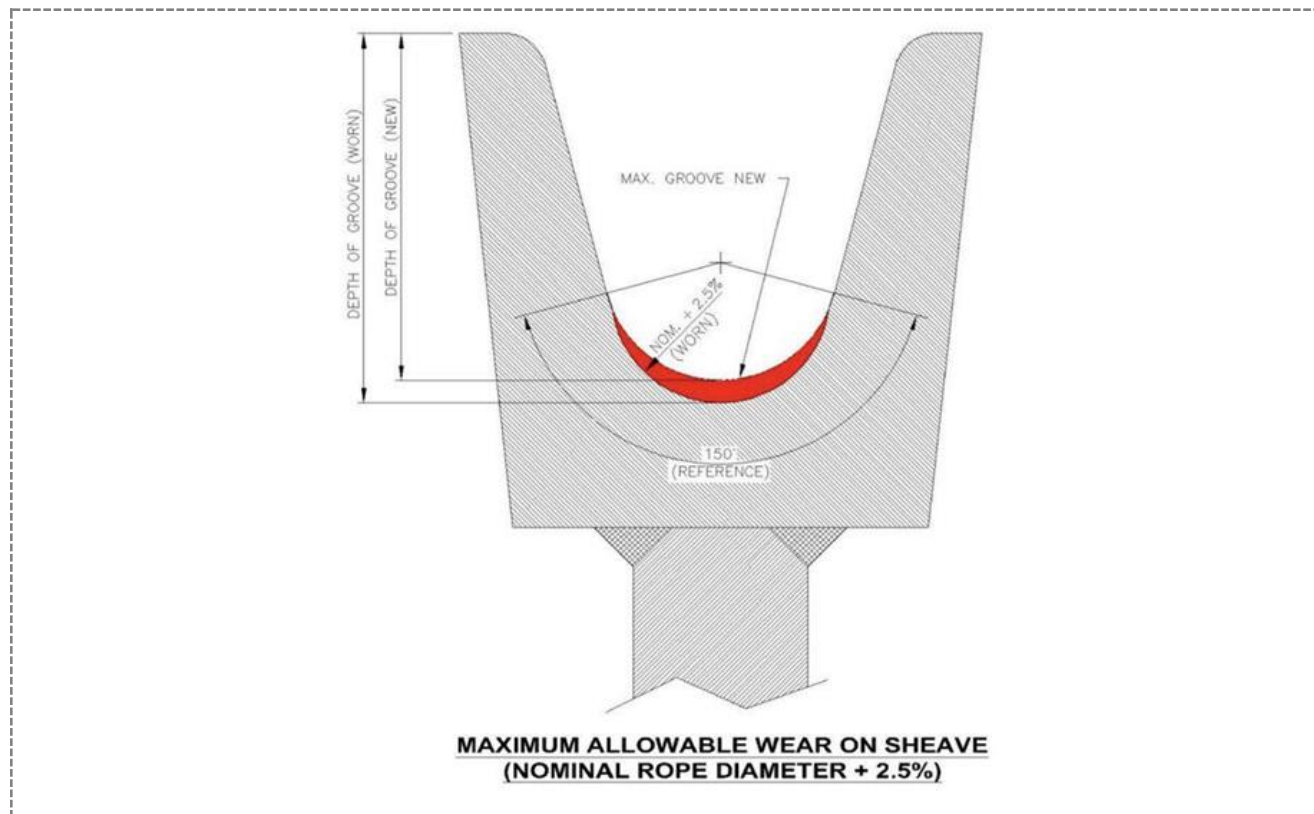


Figure 7 - CBDS-REF008 - Reference groove profile and nominal symmetrical wear pattern illustration.

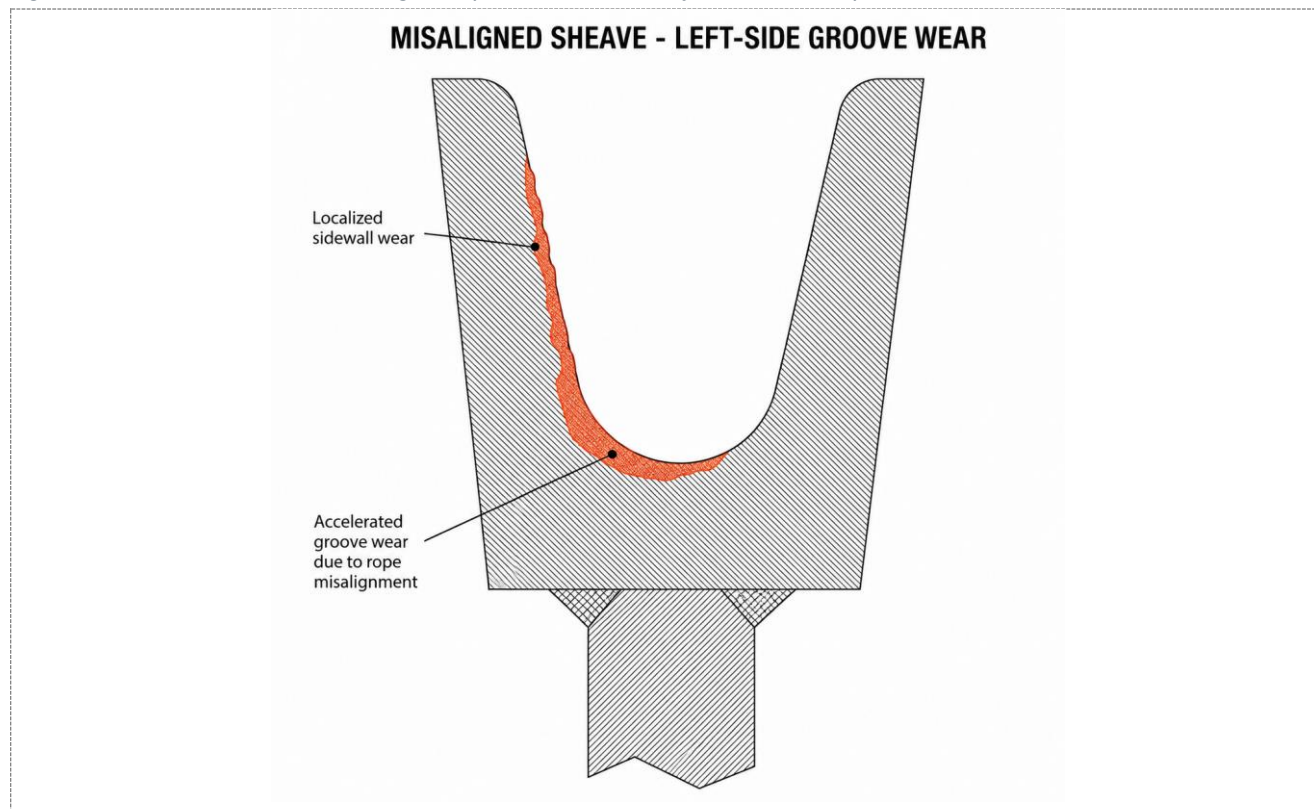


Figure 8 - CBDS-REF005 - Illustrative left-side groove wear produced by angular rope tracking.

14.5 Forces on the Sheave Bearings and Shaft

CBDS-REF007 illustrates the mechanism by which an off-center rope line can introduce unequal loading into the sheave support bearings. No bearing force was measured during this inspection; the illustration supports the recommendation for bearing/shaft condition verification during and after any geometry/alignment correction identified as necessary by the survey.

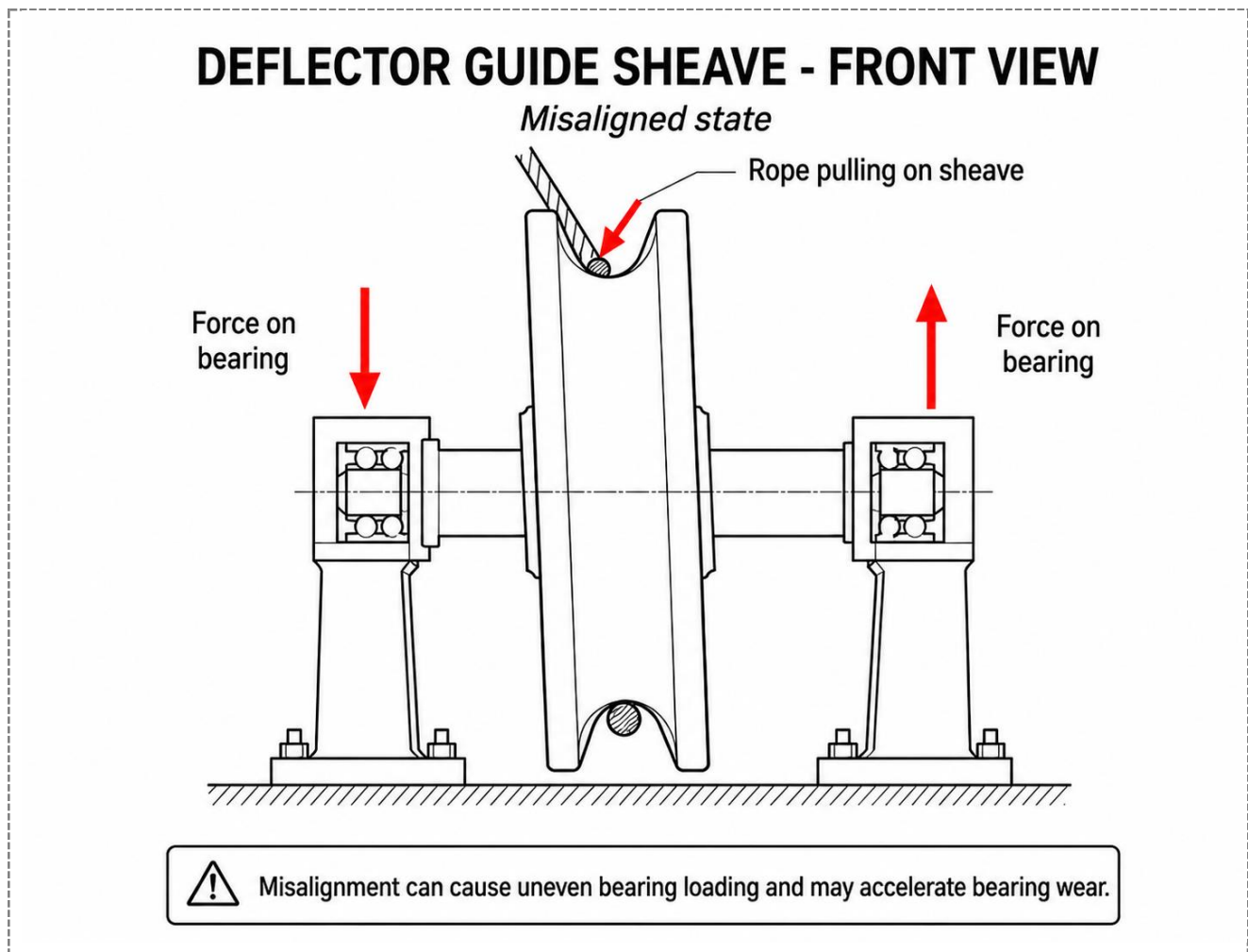


Figure 9 - CBDS-REF007 - Illustrative bearing-load mechanism associated with a misaligned sheave.

14.6 Deflector-Guide Sheave Rope-Exit Fleet Angle

The following schematic illustrations show the desired rope-exit condition at the Deflector-Guide Sheaves, an example of a possible non-zero downstream exit fleet angle, and a conceptual lower overhead guide-sheave arrangement intended to control the rope path through the Deflector-Guide Sheave exit region. The desired condition is for the rope to leave each Deflector-Guide Sheave at a fleet angle of 0° or as close to 0° as practicable. The proposed guide-sheave arrangement is conceptual only and is included to support the later system-level engineering assessment and recommendations; it is not a dimensional design, verified as-built arrangement or measured alignment survey.

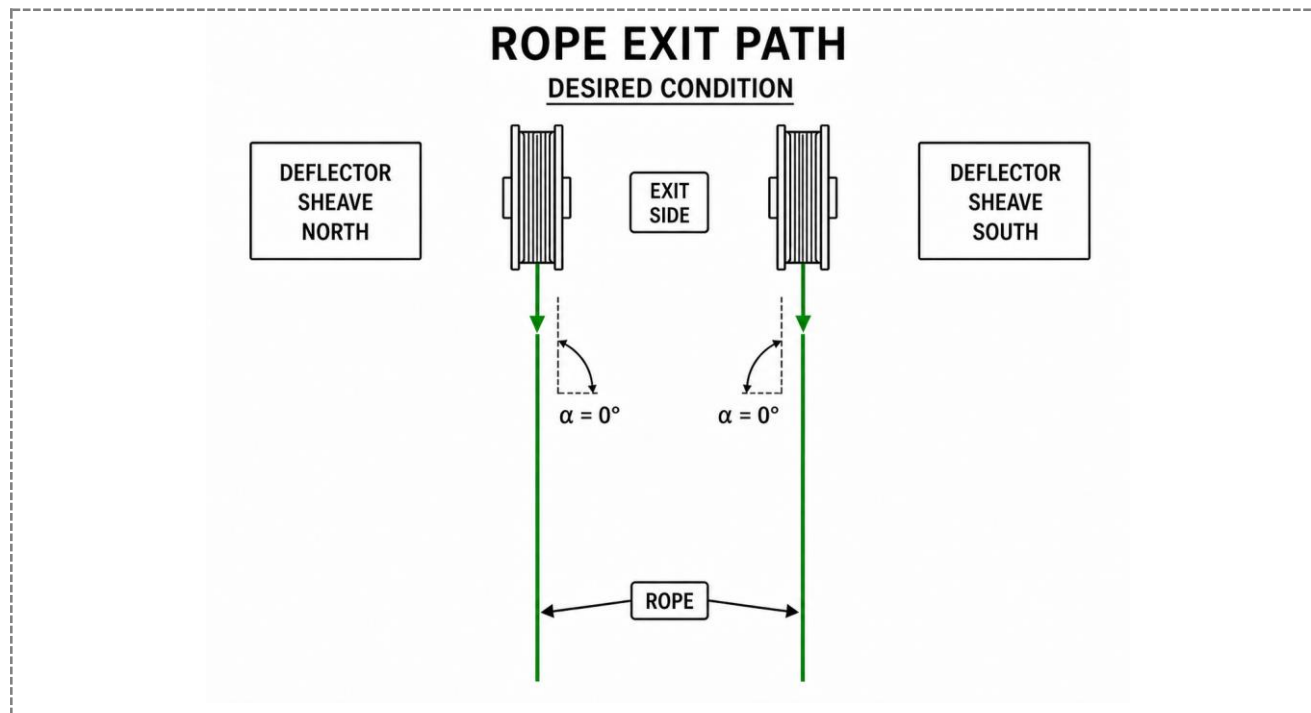


Figure 10 - CBDS-REF011 - Desired fleet angle of the rope exiting the Deflector-Guide Sheaves.

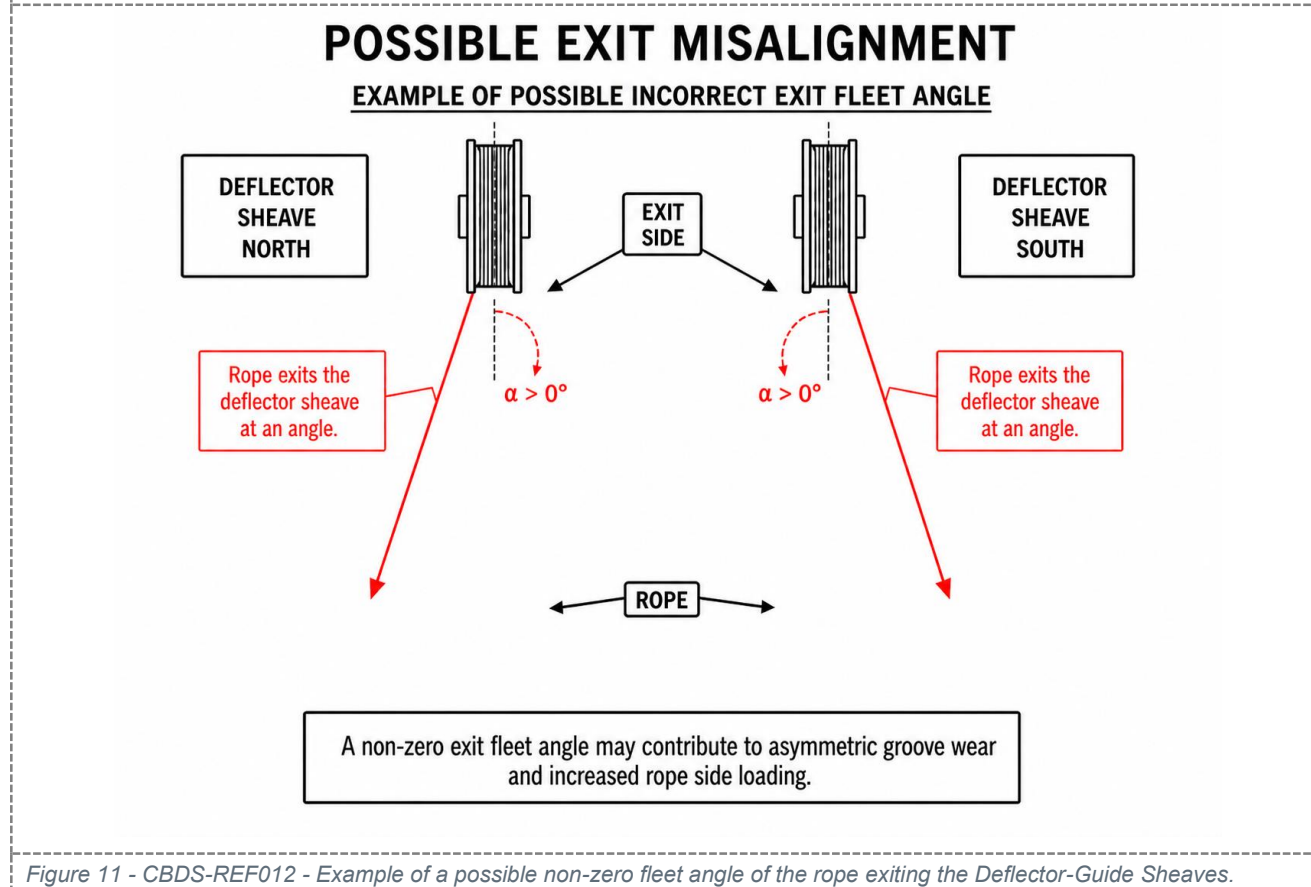


Figure 11 - CBDS-REF012 - Example of a possible non-zero fleet angle of the rope exiting the Deflector-Guide Sheaves.

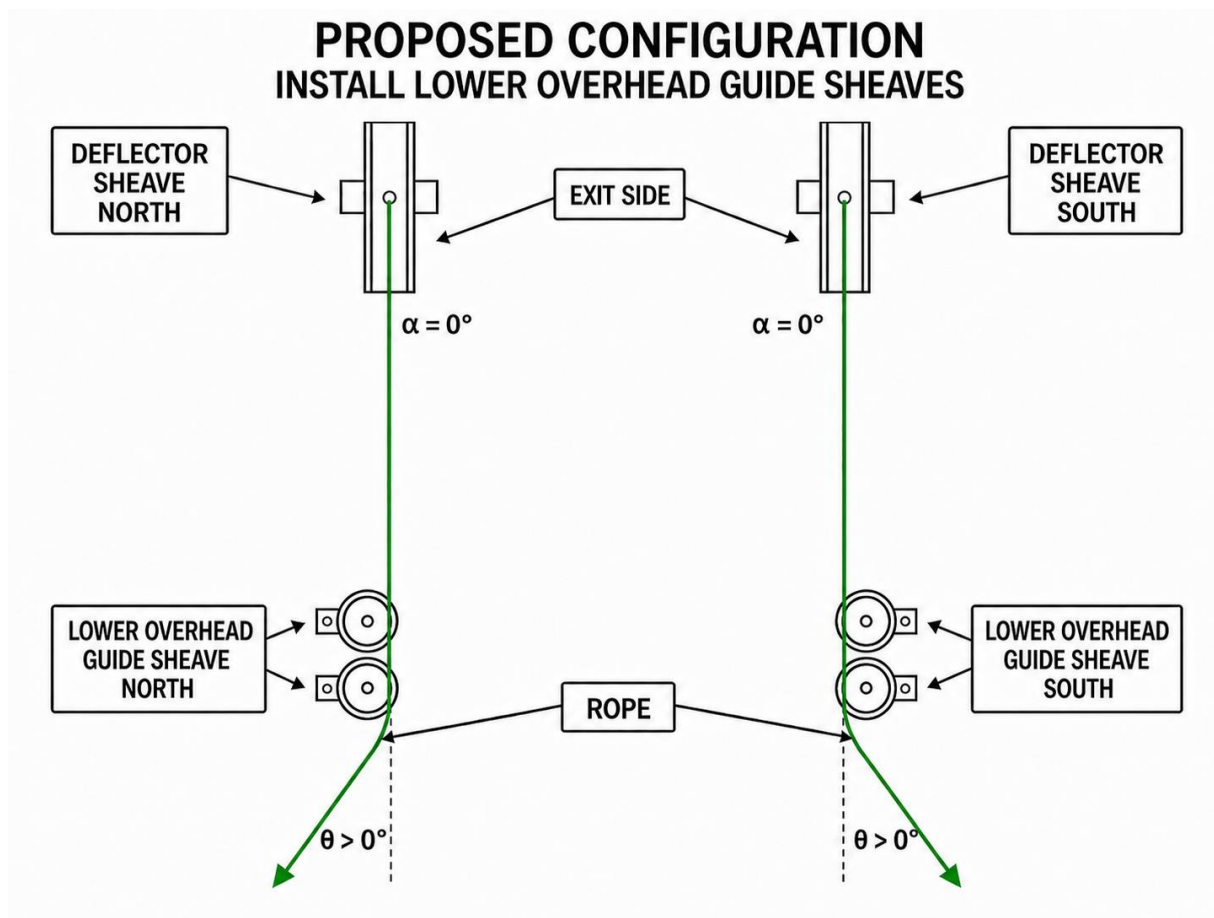


Figure 12 - CBDS-REF013 - Proposed introduction of lower overhead guide sheaves to manage the exit fleet angle from the Deflector-Guide Sheaves.

SECTION C**FINDINGS, RISK & ACTIONS**

Repeatable asset findings, defect traceability, engineering assessment and maintenance recommendations.

**15 Asset Descriptions****15.1 Drive Wheel North**

Drive Wheel North is located on the North side of the Wessels Mine cable belt main drive arrangement and drives/guides the North cable belt rope through the main drive station. The supplied inspection photographs are used to document rope tracking and groove condition; detailed OEM identification and bearing data were not supplied.

Table 25 - Drive Wheel North - asset description summary

Field	Entry
Asset	Drive Wheel North
Operational duty	drives/guides the North cable belt rope through the main drive station
Location	Cable Belt Main Drive - North
Principal components	Drive wheel, segmented groove/liner assembly, shaft, bearings, brake assemblies and supports
Reference drawing / figure	CBDS-REF001; PHOTO-DWN series
Inspection access state	Not recorded in supplied compilation notes

15.2 Drive Wheel South

Drive Wheel South is located on the South side of the Wessels Mine cable belt main drive arrangement and drives/guides the South cable belt rope through the main drive station. The supplied inspection photographs are used to document rope tracking and groove condition; detailed OEM identification and bearing data were not supplied.

Table 26 - Drive Wheel South - asset description summary

Field	Entry
Asset	Drive Wheel South
Operational duty	drives/guides the South cable belt rope through the main drive station
Location	Cable Belt Main Drive - South
Principal components	Drive wheel, groove/liner assembly, shaft, bearings, brake assemblies and supports
Reference drawing / figure	CBDS-REF001; PHOTO-DWS series
Inspection access state	Not recorded in supplied compilation notes

15.3 Deflector-Guide Sheave North

Deflector-Guide Sheave North is located on the North side of the Wessels Mine cable belt main drive arrangement and redirects the North cable belt rope after it exits the Drive Wheel North. The supplied inspection photographs are used to document rope tracking and groove condition; detailed OEM identification and bearing data were not supplied.

Table 27 - Deflector-Guide Sheave North - asset description summary

Field	Entry
Asset	Deflector-Guide Sheave North
Operational duty	redirects the North cable belt rope after it exits the Drive Wheel North
Location	Cable Belt Main Drive - North
Principal components	Deflector-guide sheave, rope groove, shaft, bearings and support structure
Reference drawing / figure	CBDS-REF001; PHOTO-DGN series
Inspection access state	Not recorded in supplied compilation notes

15.4 Deflector-Guide Sheave South

Deflector-Guide Sheave South is located on the South side of the Wessels Mine cable belt main drive arrangement and redirects the South cable belt rope after it exits the Drive Wheel South. The supplied inspection photographs are used to document rope tracking and groove condition; detailed OEM identification and bearing data were not supplied.

Table 28 - Deflector-Guide Sheave South - asset description summary

Field	Entry
Asset	Deflector-Guide Sheave South
Operational duty	redirects the South cable belt rope after it exits the Drive Wheel South
Location	Cable Belt Main Drive - South
Principal components	Deflector-guide sheave, rope groove, shaft, bearings and support structure
Reference drawing / figure	CBDS-REF001; PHOTO-DGS series
Inspection access state	Not recorded in supplied compilation notes

16 Inspection Findings

16.1 Drive Wheel North

Defect Ref ID	DEF-001	Asset	Drive Wheel North
Finding title	Off-center rope departure with asymmetric groove/liner deterioration requiring sheave-pair geometry/alignment verification.	Location	Cable Belt Main Drive - North

16.1.1 Asset Overview and General Condition

Drive Wheel North | Cable Belt Main Drive - North | Overall condition: Poor | Priority engineering review required. The condition rating reflects visible off-center rope departure and localized asymmetric deterioration of the segmented groove/liner assembly within the rope-exit region. No critical fracture or catastrophic component failure is established by the supplied photographic evidence; however, the observed contact and wear mechanism requires corrective investigation and verification.

16.1.2 Inspection Observation

PHOTO-DWN-001 and PHOTO-DWN-002 show the North cable-belt rope displaced laterally relative to the apparent Drive Wheel groove centerline at the rope-exit region. The offset is visible from opposing viewpoints. The rule visible in PHOTO-DWN-001 provides contextual scale only; no formal alignment measurement is recorded.

PHOTO-DWN-003 and PHOTO-DWN-004 document the condition of the segmented groove/liner assembly at opposing sides of the rope-exit region. Localized edge deterioration and material loss are visible, with a pronounced difference in condition between the two sides of the groove. The wear/damage distribution is therefore visibly asymmetric.

No remaining liner thickness, groove-depth measurement, dimensional profile measurement or verified OEM/client liner discard criterion was supplied.

16.1.3 Rope Groove and Rim Condition

The segmented groove/liner assembly exhibits localized deterioration within the rope-exit region. One side shows more pronounced edge deformation, material loss and localized wear than the opposing side, which remains comparatively intact. The resulting wear pattern is not symmetrical about the apparent groove centerline.

The visible condition indicates that rope contact is not being distributed uniformly across the intended groove profile. The supplied evidence does not establish the remaining liner thickness, absolute groove depth, full circumferential extent of the damage or acceptance against a verified OEM/client discard criterion.

Groove / liner condition: Localized asymmetric deterioration | Dimensional service margin not established.

16.1.4 Rope Tracking and Alignment

Rope tracking: PHOTO-DWN-001 and PHOTO-DWN-002 show the rope departing the Drive Wheel North laterally displaced from the apparent groove centerline. The opposing views provide consistent qualitative evidence that the rope-exit path is off-center.

When considered together with the asymmetric Drive Wheel liner/groove wear and the corresponding Deflector-Guide Sheave North groove evidence, the observed condition is consistent with the rope travelling on an angular path between the paired sheaves rather than following a common sheave/rope centerline.

Alignment limitation: No formal laser, optical or dimensional alignment survey was supplied. The presence of any relative misalignment, and if present its magnitude, direction and geometric source, remain to be established by the formal survey.

Overall status: Abnormal rope tracking evident | Formal geometry/alignment verification required | Corrective action to be defined from the survey results.

16.1.5 Hub, Shaft and Bearing Housing

The supplied evidence does not establish the internal condition of the Drive Wheel North bearings or shaft. No bearing temperature, vibration, axial-position/end-float, clearance or shaft-alignment measurements were supplied.

The observed off-center rope path has the potential to introduce non-uniform lateral loading into the sheave/support system; however, no bearing load or resulting bearing/shaft damage was measured or confirmed during this inspection. Bearing and shaft condition should therefore be verified as part of the alignment-correction and follow-up assessment.

16.1.6 Structural Components, Fasteners and Guards

No material structural defect was specifically identified from the supplied evidence associated with this finding. The available photographs are not a substitute for a dedicated structural, fastener, brake-interface or NDT inspection. Accessible supports, mounting fasteners, guards and associated interfaces should be checked during the alignment-correction intervention and any additional defects recorded separately.

16.1.7 Corrosion, Coating and Housekeeping

No separate material corrosion or coating defect was quantified from the supplied evidence for this finding. Surface contamination is present within the groove/liner region and may obscure local wear boundaries. The rope-contact surfaces should be cleaned before detailed dimensional/profile assessment so that liner condition, material loss and the required renewal scope can be reliably documented.

16.1.8 Engineering Assessment

ENGINEERING INTERPRETATION

The combined Drive Wheel North evidence demonstrates two related condition indicators: lateral displacement of the rope at the Drive Wheel exit and asymmetric deterioration of the segmented groove/liner assembly. PHOTO-DWN-001 and PHOTO-DWN-002 show the rope departing laterally relative to the apparent groove centerline from opposing viewpoints, while PHOTO-DWN-003 and PHOTO-DWN-004 show a pronounced difference in localized liner/groove deterioration between the opposing sides of the rope-exit region.

The unequal wear distribution is not consistent with a nominally centered and symmetrical rope-contact pattern. When considered together with the corresponding Deflector-Guide Sheave North groove evidence, the observed condition is consistent with a possible relative geometry/alignment error between the paired sheaves that could produce an angular rope path. The presence, magnitude and direction of such an error remain to be established by formal survey.

Under this condition, the rope departs the Drive Wheel with increased lateral contact against one side of the groove/liner assembly rather than remaining centered within the intended rope-contact profile. Repeated side contact provides a credible mechanism for the localized edge deformation, material loss and asymmetric wear visible at the Drive Wheel rope-exit region. The same mechanism may also contribute to localized rope abrasion and non-uniform loading of the sheave support system.

The supplied evidence does not establish whether a geometry/alignment error is present, nor, if present, its magnitude or direction. Remaining liner thickness, absolute groove depth, full circumferential extent of deterioration and acceptance against a verified OEM/client liner-wear criterion are also not established. No conclusion regarding bearing or shaft damage can be made from the available evidence.

On the evidence presently available, the Drive Wheel North condition is therefore considered to require priority corrective investigation. A formal alignment survey of the Drive Wheel North / Deflector-Guide Sheave North pair should be performed using a documented reference datum, after which the relative sheave alignment should be corrected as required. The groove/liner assembly should be cleaned and dimensionally/profile assessed to document the as-found condition and establish the pre-renewal baseline. The Drive Wheel North liner assembly should then be renewed as part of the coordinated corrective intervention defined in Section 20.

Following correction, the formal survey and any corrective actions, the rope path should be rechecked under the applicable inspection condition and post-correction photographs/measurements retained as the new alignment and condition baseline.

Condition classification: Poor | Level 3 - Significant | P2 - Urgent.

16.1.9 Defect Classification and Recommended Action*Table 29 - Drive Wheel North finding classification and action*

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-001	Poor	Level 3 - Significant	P2 - Urgent	Survey the Drive Wheel North / Deflector-Guide Sheave North alignment to a recorded datum and correct as required. Clean and dimensionally/profile assess the existing Drive Wheel North groove/liner to document the as-found condition and establish the pre-renewal baseline, then renew the Drive Wheel North liner assembly as part of the coordinated corrective intervention defined in Section 20. Inspect the North 45 mm rope for localized side-contact damage	Mechanical / alignment / rope specialist	Alignment survey/correction record; post-correction rope-tracking evidence; groove/liner measurements against verified criteria; rope inspection and close-out record.

16.1.10 Photographic Evidence Sheet



Photo ID
 PHOTO-DWN-001

Finding ID
 DEF-001

Asset
 Drive Wheel North

Component / location
 Rope exit / groove

View / direction
 Left-side contextual view of rope exit

Date / time
 18/07/2026 (inspection date)

Caption
 Left-side view showing the rope exiting Drive Wheel North off the groove centerline.

Figure 13 - Drive Wheel North evidence - PHOTO-DWN-001 / DEF-001

Table 30 - Photographic evidence assessment - PHOTO-DWN-001

Observation	Engineering assessment	Recommended action
The North rope is visibly positioned laterally relative to the apparent groove centerline at the Drive Wheel exit. The rule provides contextual scale only; no formal alignment value is recorded.	The photograph provides qualitative evidence of an apparent off-center rope-exit path. Together with the paired Deflector-Guide Sheave North evidence, the condition warrants verification of the sheave-pair geometry; any alignment error remains unquantified.	Include this location in the formal North sheave-pair alignment survey. Recheck the rope position after correction and inspect the rope for localized side-contact damage.

16.1.11 Photographic Evidence Sheet

**Photo ID**
PHOTO-DWN-002**Finding ID**
DEF-001**Asset**
Drive Wheel North**Component / location**
Rope exit / groove**View / direction**
Opposing contextual view of rope exit**Date / time**
18/07/2026 (inspection date)**Caption**

Opposing view showing apparent lateral offset of the rope path at the Drive Wheel North exit.

Figure 14 - Drive Wheel North evidence - PHOTO-DWN-002 / DEF-001

Table 31 - Photographic evidence assessment - PHOTO-DWN-002

Observation	Engineering assessment	Recommended action
The opposing view again shows the rope displaced laterally at the Drive Wheel North exit rather than centered within the apparent groove path.	This opposing view corroborates PHOTO-DWN-001 and strengthens the qualitative evidence of off-center rope departure. The presence of any relative misalignment, and if present its magnitude, direction and geometric source, remain to be established by formal survey.	Confirm the Drive Wheel North and Deflector-Guide Sheave North centerline/shaft relationship by formal survey, correct the relative alignment as required, and capture post-correction tracking photographs.

16.1.12 Photographic Evidence Sheet

Photo ID
PHOTO-DWN-003

Finding ID
DEF-001

Asset
Drive Wheel North

Component / location
Groove / liner - right-hand side

View / direction
Close-up of liner / groove damage

Date / time
18/07/2026 (inspection date)

Caption

Localized deterioration and material loss on one side of the segmented groove/liner assembly at the Drive Wheel North rope exit. The condition is consistent with concentrated lateral rope contact associated with the observed off-center rope path.

Figure 15 - Drive Wheel North evidence - PHOTO-DWN-003 / DEF-001

Table 32 - Photographic evidence assessment - PHOTO-DWN-003

Observation	Engineering assessment	Recommended action
Localized deterioration, edge deformation and material loss are visible on one side of the segmented groove/liner assembly within the rope-exit region.	The localized damage is consistent with concentrated lateral side contact associated with the off-center rope path. When considered with the opposing-side condition and the paired Deflector-Guide Sheave North evidence, the damage supports the assessment of an asymmetric contact mechanism. Remaining liner thickness and acceptance against verified OEM/client criteria are not established by this photograph.	After alignment verification/correction, clean and dimensionally/profile assess the affected liner/groove region to document the as-found condition and establish the pre-renewal baseline, then renew the Drive Wheel North liner assembly as part of the coordinated corrective intervention.

16.1.13 Photographic Evidence Sheet



Photo ID
 PHOTO-DWN-004

Finding ID
 DEF-001

Asset
 Drive Wheel North

Component / location
 Groove / liner - opposing side

View / direction
 Opposing-side close-up of liner / groove

Date / time
 18/07/2026 (inspection date)

Caption

Opposing side of the Drive Wheel North groove showing comparatively limited sleeve-insert deterioration. The contrast with PHOTO-DWN-003 demonstrates an asymmetric liner/groove wear distribution at the rope-exit region.

Figure 16 - Drive Wheel North evidence - PHOTO-DWN-004 / DEF-001

Table 33 - Photographic evidence assessment - PHOTO-DWN-004

Observation	Engineering assessment	Recommended action
The opposing side of the Drive Wheel North groove/liner exhibits substantially less visible edge deterioration and material loss than the side documented in PHOTO-DWN-003. The liner edges remain comparatively intact, with only limited localized wear visible in the rope-exit region.	The pronounced difference in condition between the two sides provides additional evidence of an asymmetric rope-contact pattern rather than nominally centered and symmetrical groove loading. In conjunction with the observed lateral rope offset, the wear distribution is consistent with angular rope departure between the paired North sheaves.	After alignment verification/correction, clean and dimensionally/profile assess both sides of the Drive Wheel North groove/liner to document the as-found condition, retain the opposing-side photographs as baseline evidence, and renew the Drive Wheel North liner assembly as part of the coordinated corrective intervention.

16.2 Drive Wheel South

Defect Ref ID	DEF-002	Asset	Drive Wheel South
Finding title	Asymmetric Drive Wheel South groove/liner deterioration consistent with angular rope tracking and possible rope-entry fleet-angle offset	Location	Cable Belt Main Drive - South

16.2.1 Asset Overview and General Condition

Drive Wheel South | Cable Belt Main Drive - South | Overall condition: Poor | Priority engineering review required. The condition rating reflects visible localized asymmetric deterioration of the segmented groove/liner assembly, with the more pronounced damage recorded on the left-hand side and comparatively lower visible wear on the right-hand side. No critical fracture or catastrophic component failure is established by the supplied photographic evidence; however, the observed contact and wear mechanism requires corrective investigation and verification.

16.2.2 Inspection Observation

PHOTO-DWS-001 and PHOTO-DWS-002 show unequal wear between the two sides of the Drive Wheel South groove/liner. The left-hand side is the more pronounced damage region, while the right-hand side shows comparatively lower visible damage. The wear/damage distribution is therefore visibly asymmetric and is a mirror-image of the North drive wheel condition.

PHOTO-DWS-003 provides rear/bearing and brake-system context only and does not, by itself, establish the wear mechanism.

No remaining liner thickness, groove-depth measurement, dimensional profile measurement or verified OEM/client liner discard criterion was supplied.

16.2.3 Rope Groove and Rim Condition

The segmented groove/liner assembly exhibits localized deterioration within the rope-entry/working contact region. The left-hand side shows more pronounced edge deformation, material loss and localized wear, while the opposing right-hand side remains comparatively intact. The wear pattern is therefore a mirror-image of the North drive wheel condition and is not symmetrical about the apparent groove centerline.

The visible condition indicates that rope contact is not being distributed uniformly across the intended groove profile. The supplied evidence does not establish the remaining liner thickness, absolute groove depth, full circumferential extent of the damage or acceptance against a verified OEM/client discard criterion.

Groove / liner condition: Localized asymmetric deterioration | Left-hand side more severely affected | Dimensional service margin not established.

16.2.4 Rope Tracking and Alignment

Rope tracking: The visible left/right wear contrast shows sustained non-uniform rope contact across the Drive Wheel South groove/liner. When considered together with the corresponding Deflector-Guide Sheave South groove evidence, the observed condition is consistent with the rope travelling on an angular path between the paired sheaves rather than following a common sheave/rope centerline.

A broader system-level consideration is that the mirrored damage pattern across the North and South drive wheels may also be qualitatively consistent with a non-zero rope-entry fleet angle at the top of the drive-wheel pair, as illustrated schematically in CBDS-REF009 and CBDS-REF010. This possible mechanism is not confirmed by the present inspection evidence and should be evaluated as part of the integrated alignment review of the drive sheaves, Deflector-Guide Sheaves and upper guide sheave arrangement.

Alignment limitation: No formal laser, optical or dimensional alignment survey was supplied. The presence of any relative sheave-pair misalignment or rope-entry fleet-angle offset, and if present its magnitude, direction and geometric source, remain to be established by the formal survey.

Overall status: Abnormal contact pattern consistent with angular rope tracking | Formal geometry/alignment verification required | Corrective action to be defined from the survey results.

16.2.5 Hub, Shaft and Bearing Housing

The supplied evidence does not establish the internal condition of the Drive Wheel South bearings or shaft. No bearing temperature, vibration, axial-position/end-float, clearance or shaft-alignment measurements were supplied.

The observed asymmetric contact pattern has the potential to introduce non-uniform lateral loading into the sheave/support system; however, no bearing load or resulting bearing/shaft damage was measured or confirmed during this inspection. Bearing and shaft condition should therefore be verified as part of the alignment-correction and follow-up assessment.

16.2.6 Structural Components, Fasteners and Guards

No material structural defect was specifically identified from the supplied evidence associated with this finding. The available photographs are not a substitute for a dedicated structural, fastener, brake-interface or NDT inspection. Accessible supports, mounting fasteners, guards and associated interfaces should be checked during the alignment-correction intervention and any additional defects recorded separately.

16.2.7 Corrosion, Coating and Housekeeping

No separate material corrosion or coating defect was quantified from the supplied evidence for this finding. Surface contamination is present within the groove/liner region and may obscure local wear boundaries. The rope-contact surfaces should be cleaned before detailed dimensional/profile assessment so that liner condition, material loss and the required renewal scope can be reliably documented.

16.2.8 Engineering Assessment

ENGINEERING INTERPRETATION

The Drive Wheel South evidence demonstrates localized asymmetric deterioration of the segmented groove/liner assembly, with the left-hand side more severely affected than the opposing right-hand side. The unequal wear distribution is not consistent with a nominally centered and symmetrical rope-contact pattern.

When considered together with the corresponding Deflector-Guide Sheave South groove evidence, the observed condition is consistent with a possible relative geometry/alignment error between the paired sheaves that could produce an angular rope path. Repeated side contact provides a credible mechanism for the localized edge deformation, material loss and asymmetric wear visible at the Drive Wheel South rope-contact region. The same mechanism may also contribute to localized rope abrasion and non-uniform loading of the sheave support system.

A broader system-level consideration is that the mirrored left/right damage relationship between the South and North drive wheels may also be qualitatively consistent with the possibility that the rope enters the drive-wheel pair at a non-zero fleet angle, as illustrated schematically in CBDS-REF009 and CBDS-REF010. This broader system-level mechanism has not been measured or confirmed in the present inspection and should be considered during the integrated review of the drive sheaves, Deflector-Guide Sheaves and upper guide sheave arrangement.

The supplied evidence does not establish whether a geometry/alignment error or rope-entry fleet-angle offset is present, nor, if present, its magnitude or direction. Remaining liner thickness, absolute groove depth, full circumferential extent of deterioration and acceptance against a verified OEM/client liner-wear criterion are also not established. No conclusion regarding bearing or shaft damage can be made from the available evidence.

On the evidence presently available, the Drive Wheel South condition is therefore considered to require priority corrective investigation. A formal alignment survey of the Drive Wheel South / Deflector-Guide Sheave South pair should be performed using a documented reference datum, after which the relative sheave alignment should be corrected as required. As part of the broader system review, the rope-entry path into the Drive Wheel South should also be assessed to determine whether a non-zero fleet angle is contributing to the observed contact pattern. The groove/liner assembly should be cleaned and dimensionally/profile assessed to document the as-found condition and establish the pre-renewal baseline. The Drive Wheel South liner assembly should then be renewed as part of the coordinated corrective intervention defined in Section 20.

The South 45 mm cable-belt rope should also be inspected in the corresponding contact region for localized abrasion, wire damage, distortion or other evidence of sustained side contact. Following correction, the rope path should be rechecked under the applicable inspection condition and post-correction photographs/measurements retained as the new alignment and condition baseline.

Condition classification: Poor | Level 3 – Significant | P2 – Urgent.

16.2.9 Defect Classification and Recommended Action*Table 34 - Drive Wheel South finding classification and action*

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-002	Poor	Level 3 – Significant	P2 – Urgent	Perform a formal alignment survey of the Drive Wheel South / Deflector-Guide Sheave South pair using a documented datum and correct the relative sheave alignment as required. As part of the broader system review, assess whether the rope-entry fleet angle into Drive Wheel South is non-zero. Clean and dimensionally/profile assess the existing Drive Wheel South groove/liner to document the as-found condition and establish the pre-renewal baseline, then renew the Drive Wheel South liner assembly as part of the coordinated corrective intervention defined in Section 20. Inspect the South 45 mm cable-belt rope for localized side-contact damage.	Mechanical / alignment / rope specialist	Recorded alignment survey and correction report; post-correction rope-tracking evidence; groove/liner dimensional/profile assessment against verified criteria; rope inspection record and close-out evidence.

16.2.10 Photographic Evidence Sheet



Photo ID
PHOTO-DWS-001

Finding ID
DEF-002

Asset
Drive Wheel South

Component / location
Groove / liner - right-hand side

View / direction
Right-hand groove / rope view

Date / time
18/07/2026 (inspection date)

Caption
Right-hand side of Drive Wheel South showing comparatively lower visible wear.

Figure 17 - Drive Wheel South evidence - PHOTO-DWS-001 / DEF-002

Table 35 - Photographic evidence assessment - PHOTO-DWS-001

Observation	Engineering assessment	Recommended action
The right-hand side of the Drive Wheel South groove/liner shows comparatively lower visible deterioration than the opposing left-hand side. This view represents the less severely affected side of the asymmetric contact pattern.	The lower wear on this side forms part of a sustained asymmetric left/right contact pattern rather than a uniform centered wear condition. The contrast with PHOTO-DWS-002 supports the interpretation that rope contact is being concentrated preferentially toward the opposing side of the groove/liner.	Measure both sides of the Drive Wheel South groove/liner after the South sheave-pair alignment is surveyed and corrected, and establish a post-correction baseline profile for future condition comparison.

16.2.11 Photographic Evidence Sheet

**Photo ID**

PHOTO-DWS-002

Finding ID

DEF-002

Asset

Drive Wheel South

Component / location

Groove / liner - left-hand side

View / direction

Left-hand groove / rope view

Date / time

18/07/2026 (inspection date)

Caption

Left-hand side of Drive Wheel South showing the more pronounced asymmetric wear/contact region.

Figure 18 - Drive Wheel South evidence - PHOTO-DWS-002 / DEF-002

Table 36 - Photographic evidence assessment - PHOTO-DWS-002

Observation	Engineering assessment	Recommended action
The left-hand side of the Drive Wheel South groove/liner shows more pronounced visible wear, localized deterioration and contact damage than the opposing right-hand side.	The contrast between the two sides supports a sustained asymmetric contact pattern consistent with an angular rope path between the paired South sheaves. In the broader system context, the mirrored damage pattern relative to the North drive wheel may also be qualitatively consistent with a non-zero rope-entry fleet angle into the drive-wheel pair, although that mechanism is not confirmed by this photograph alone.	Following the survey, correct the South sheave-pair geometry where a non-compliant alignment/fleet-angle condition is confirmed. Dimensionally/profile assess the existing Drive Wheel South groove/liner to document the as-found condition and establish the pre-renewal baseline, then renew the Drive Wheel South liner assembly as part of the coordinated corrective intervention. Include assessment of the rope-entry path into Drive Wheel South in the broader system review.

16.3 Deflector-Guide Sheave North

Defect Ref ID	DEF-003	Asset	Deflector-Guide Sheave North
Finding title	Off-center groove contact and asymmetric wear consistent with angular rope entry and possible non-zero exit fleet angle	Location	Cable Belt Main Drive - North

16.3.1 Asset Overview and General Condition

Deflector-Guide Sheave North | Cable Belt Main Drive - North | Overall condition: Poor | Priority engineering review required. The condition rating reflects visible off-center groove contact and localized asymmetric deterioration within the groove/rope-contact region. No critical fracture or catastrophic component failure is established by the supplied photographic evidence; however, the observed contact and wear mechanism requires corrective investigation and verification.

16.3.2 Inspection Observation

PHOTO-DGN-002 and PHOTO-DGN-003 show the groove profile gauge seated against a visibly non-uniform groove. The marked callouts indicate the approximate groove center/reference position and the area where the wear/contact profile would be expected if the rope were running centrally. The visible operating contact band is displaced from that indicated centered region.

Inspection field notes record an as-found groove depth of 65 mm and approximately 3 mm of local groove wear on the Deflector-Guide Sheave North. The approximately 3 mm wear location is indicated in PHOTO-DGN-002 (Figure 19). No photograph of the measuring operation was retained; however, the handwritten 65 mm groove-depth record on the sheave is visible in PHOTO-DGN-004 (Figure 21), providing additional traceability to the field note.

PHOTO-DGN-004 also shows localized polishing/wear on one side of the groove. PHOTO-DGN-005 and PHOTO-DGN-006 provide additional marked views showing lateral displacement of the worn/contact region.

The evidence is consistent with asymmetric groove wear and profile offset. The recorded 65 mm groove depth and approximately 3 mm local wear are as-found field measurements. The Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurement; however, the measurement method/instrument used for the approximate 3 mm local-wear value was not recorded. Verified nominal/new groove geometry, remaining wear allowance, external alignment measurements and OEM/client discard limits were not supplied.

16.3.3 Rope Groove and Rim Condition

The Deflector-Guide Sheave North groove exhibits pronounced asymmetric sidewall wear and an off-center contact profile. The marked profile-gauge photographs identify the approximate groove center/reference position and show that the visible wear/contact region is displaced from the intended centered contact zone. The groove wear is therefore not symmetrical about the apparent groove centerline.

Inspection field notes record an as-found groove depth of 65 mm and approximately 3 mm local groove wear. These values quantify the observed condition at the recorded inspection location but do not, by themselves, establish remaining wear allowance or acceptance because verified nominal/new groove geometry and OEM/client discard criteria were not supplied. The full circumferential extent and dimensional profile of the groove also remain to be mapped.

Groove condition: Pronounced asymmetric wear and profile offset | Recorded groove depth: 65 mm | Approximate local wear: ~3 mm | Remaining service margin against OEM/client criteria not established.

16.3.4 Rope Tracking and Alignment

Rope tracking: The marked groove-profile evidence in PHOTO-DGN-002 and PHOTO-DGN-003, together with the additional groove views in PHOTO-DGN-004 to PHOTO-DGN-006, shows that the operating contact band is displaced from the indicated centered groove/contact region. When considered together with the off-center rope departure and asymmetric wear observed at Drive Wheel North, the condition is consistent with the possibility that the rope enters the Deflector-Guide Sheave North at an angle from Drive Wheel North, as illustrated schematically in CBDS-REF002 and CBDS-REF003, rather than following the desired centered relationship shown in CBDS-REF004.

A further possible contributing factor is that, once the rope exits the Deflector-Guide Sheave North, the exit fleet angle on the downstream side may also be non-zero because no smaller guide sheave is presently identified on the exit side to control the rope path. This possible mechanism is not confirmed by the present evidence and should be verified using external measurements and alignment tools.

Alignment limitation: No formal laser, optical or dimensional alignment survey was supplied. The presence of any entry or exit rope-path geometry/alignment error, and if present its magnitude, direction and geometric source, remain to be established by the formal survey.

Overall status: Abnormal groove-contact pattern consistent with possible angular rope entry and possible non-zero exit fleet angle | Formal geometry/alignment verification required | Corrective action to be defined from the survey results.

16.3.5 Hub, Shaft and Bearing Housing

The supplied evidence does not establish the internal condition of the Deflector-Guide Sheave North bearings or shaft. No bearing temperature, vibration, axial-position/end-float, clearance or shaft-alignment measurements were supplied.

The observed off-center contact pattern has the potential to introduce non-uniform lateral loading into the sheave/support system; however, no bearing load or resulting bearing/shaft damage was measured or confirmed during this inspection. Bearing and shaft condition should therefore be verified as part of the alignment-correction and follow-up assessment.

16.3.6 Structural Components, Fasteners and Guards

No material structural defect was specifically identified from the supplied evidence associated with this finding. The available photographs are not a substitute for a dedicated structural, fastener, brake-interface or NDT inspection. Accessible supports, mounting fasteners, guards and associated interfaces should be checked during the alignment-correction intervention and any additional defects recorded separately.

16.3.7 Corrosion, Coating and Housekeeping

No separate material corrosion or coating defect was quantified from the supplied evidence for this finding. Surface contamination is present within the groove region and may obscure local wear boundaries. The rope-contact surfaces should be cleaned before detailed dimensional/profile assessment so that the as-found groove condition and material loss can be reliably documented and retained as the pre-replacement baseline.

16.3.8 Engineering Assessment

ENGINEERING INTERPRETATION

The Deflector-Guide Sheave North evidence demonstrates a pronounced off-center groove-contact condition. The marked profile-gauge photographs show that the visible wear/contact band is displaced from the approximate groove center/reference position, while the additional surface-detail and marked views show a localized worn/polished region that is not symmetrical about the intended groove profile.

Inspection field notes record an as-found groove depth of 65 mm and approximately 3 mm local groove wear. The approximately 3 mm wear is indicated in PHOTO-DGN-002 (Figure 19), and the handwritten 65 mm groove-depth value is visible on the sheave in PHOTO-DGN-004 (Figure 21). These readings provide quantitative as-found condition evidence. The Bosch PLR 50, equipment ID 009741746, is recorded as the instrument used for the groove-depth measurement; however, no photograph of the measuring operation was retained, and the measurement method/instrument used for the approximate 3 mm local-wear value was not recorded. No verified nominal/new groove geometry or OEM/client discard criterion was supplied. The remaining wear allowance therefore cannot be calculated from these values alone.

When considered together with the Drive Wheel North findings, the observed condition is consistent with the possibility that the rope enters the Deflector-Guide Sheave North on an angular path from Drive Wheel North rather than along a common centered sheave/rope centerline. Repeated off-center rope entry provides a credible mechanism for the localized sidewall wear, asymmetric polishing and displaced operating profile observed in the groove.

A further system-level possibility is that, after passing through the Deflector-Guide Sheave North, the rope may also be exiting the sheave with a non-zero fleet angle on the downstream side because no smaller exit-side guide sheave is presently identified to control the rope path. This possible mechanism has not been measured or confirmed by the present inspection evidence. It should be considered as a potential contributing factor and verified as part of the integrated review of the drive sheaves, Deflector-Guide Sheaves and downstream rope path

arrangement.

The wear pattern can therefore be interpreted in two complementary ways: an incorrect entry and/or exit rope-path relationship could contribute to the irregular groove wear, while the irregular wear pattern itself provides qualitative evidence warranting investigation of the rope-path geometry. The field measurements quantify the as-found groove condition but do not establish whether geometric misalignment is present, nor do they establish remaining service margin against an OEM/client criterion. No conclusion regarding bearing or shaft damage can be made from the available evidence.

On the evidence presently available, the Deflector-Guide Sheave North condition is therefore considered to require priority corrective investigation. A formal alignment survey of the Drive Wheel North / Deflector-Guide Sheave North relationship and the downstream exit rope path should be performed using a documented reference datum. Before removal/replacement, the 65 mm depth and approximately 3 mm local wear should be repeated at identified/indexed locations where practicable using recorded measuring equipment, then compared with verified OEM/client criteria.

The North 45 mm cable-belt rope should also be inspected in the corresponding contact region for localized abrasion, wire damage, distortion or other evidence of sustained side contact. Following correction, the rope path and groove/contact profile should be rechecked and documented as the new baseline.

Condition classification: Poor | Level 3 - Significant | P2 - Urgent.

16.3.9 Defect Classification and Recommended Action

Table 37 - Deflector-Guide Sheave North finding classification and action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-003	Poor	Level 3 – Significant	P2 – Urgent	Perform a formal alignment survey of the Drive Wheel North / Deflector-Guide Sheave North pair and the downstream exit rope path using a documented datum, and correct the rope-path/sheave alignment where a non-compliant geometry condition is confirmed. Retain the recorded as-found groove values of 65 mm depth and approximately 3 mm local wear; repeat/verify them at identified indexed locations where practicable using recorded measuring equipment. Compare the dimensional groove profile with verified OEM/client criteria and complete the recommended Deflector-Guide Sheave replacement as part of the coordinated corrective intervention. Inspect the North 45 mm cable-belt rope for localized side-contact damage.	Mechanical / alignment / rope specialist	Recorded alignment survey/correction report; repeatable groove-depth/local-wear measurements with equipment ID and indexed location; comparison with verified OEM/client criteria; replacement identification and final groove/profile baseline; post-correction rope-tracking evidence; rope inspection and close-out record.

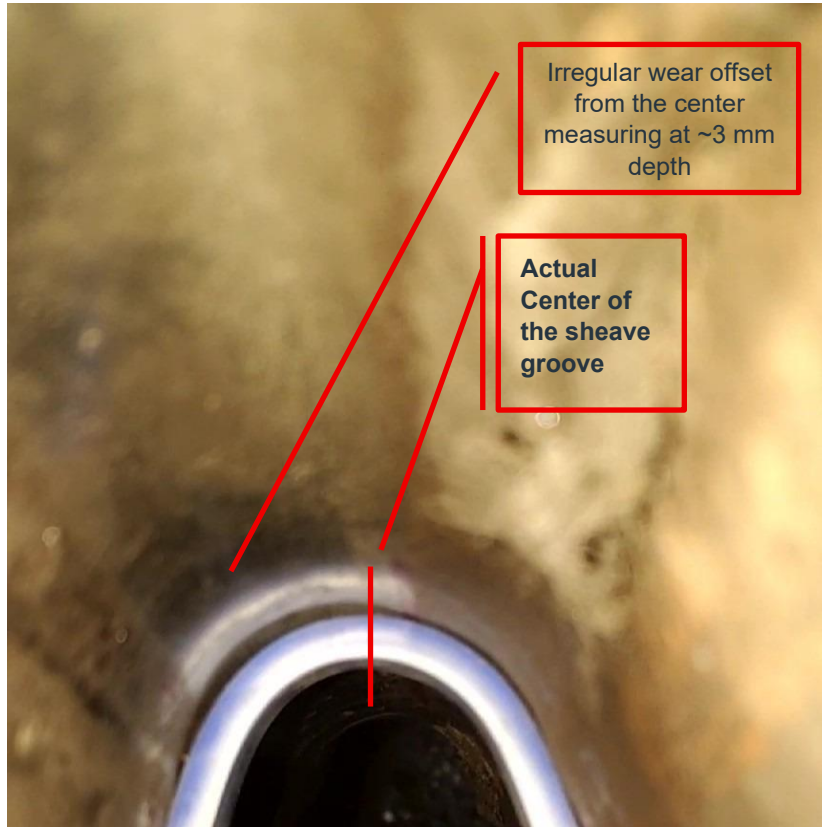
16.3.10 Photographic Evidence Sheet**Photo ID**

PHOTO-DGN-002

Finding ID

DEF-003

Asset

Deflector-Guide Sheave North

Component / location

Rope groove / profile

View / direction

Close-up profile-gauge view

Date / time

18/07/2026 (inspection date)

Caption

Profile-gauge view showing the worn Deflector-Guide Sheave North groove; field annotation/notes record approximately 3 mm local groove wear at the indicated location.

Figure 19 - Deflector-Guide Sheave North evidence - PHOTO-DGN-002 / DEF-003

Table 38 - Photographic evidence assessment - PHOTO-DGN-002

Observation	Engineering assessment	Recommended action
The profile gauge is shown seated in the Deflector-Guide Sheave North groove. The marked callouts indicate the approximate groove center/reference position and the region where centered rope contact would be expected. The visible contact/wear profile is displaced from that indicated centered region. Inspection field notes record approximately 3 mm local groove wear at the indicated location.	The marked callouts strengthen the qualitative evidence that the groove is operating off-center rather than wearing symmetrically about its intended profile. The approximately 3 mm value is an as-found field measurement; however, the measurement method/instrument used for this local-wear value was not recorded, and remaining service margin cannot be determined without verified nominal/new groove geometry and OEM/client acceptance criteria.	Retain the approximately 3 mm reading as the as-found record. Repeat it at an identified indexed position using recorded measuring equipment and compare the dimensional groove profile with verified OEM/client criteria during the corrective intervention.

16.3.11 Photographic Evidence Sheet

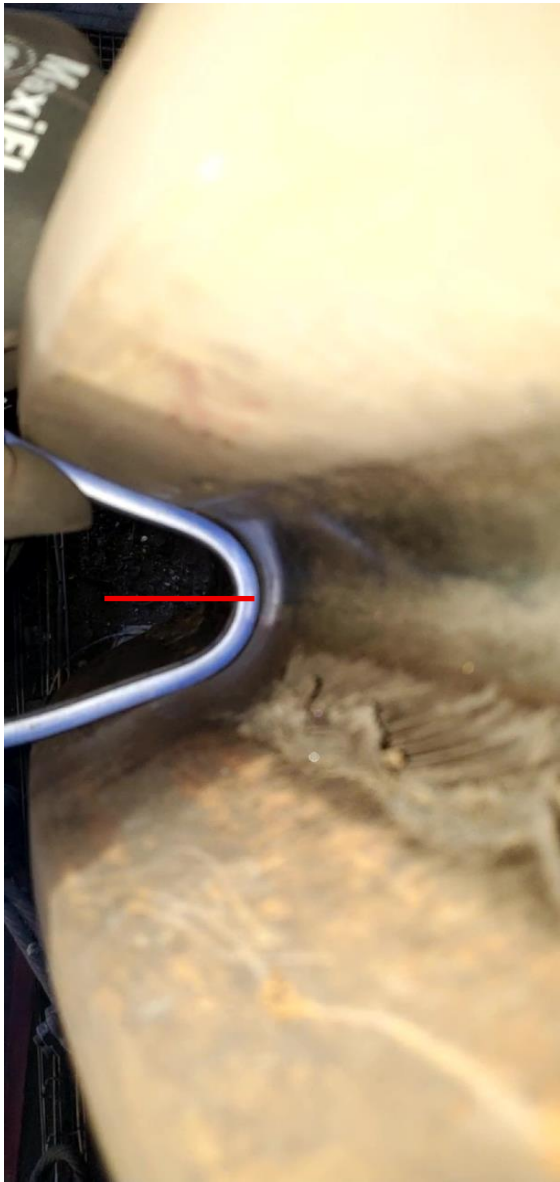


Photo ID
 PHOTO-DGN-003
Finding ID
 DEF-003
Asset
 Deflector-Guide Sheave North
Component / location
 Rope groove / profile
View / direction
 Additional close-up profile-gauge view
Date / time
 18/07/2026 (inspection date)

Caption
 Additional profile-gauge view of the worn Deflector-Guide Sheave North groove.

Figure 20 - Deflector-Guide Sheave North evidence - PHOTO-DGN-003 / DEF-003
Table 39 - Photographic evidence assessment - PHOTO-DGN-003

Observation	Engineering assessment	Recommended action
A second marked profile-gauge view again shows the indicated groove center/reference position and the expected centered contact region, while the visible operating wear/contact band remains displaced from that region.	This view corroborates PHOTO-DGN-002 and supports the conclusion that the groove profile is asymmetric and laterally offset rather than centered and uniform. The remaining wear allowance cannot be determined without dimensional survey data.	Dimensionally map the North groove profile, compare it with verified design/discard criteria for condition documentation, and retain the results as the as-found baseline before the recommended Deflector-Guide Sheave North replacement.

16.3.12 Photographic Evidence Sheet

Photo ID
PHOTO-DGN-004

Finding ID
DEF-003

Asset
Deflector-Guide Sheave North

Component / location
Rope groove sidewall

View / direction
Close-up groove surface view

Date / time
18/07/2026 (inspection date)

Caption
Groove surface detail showing asymmetric polishing/wear on Deflector-Guide Sheave North. The handwritten 65 mm groove-depth field record is visible on the sheave.

Figure 21 - Deflector-Guide Sheave North evidence - PHOTO-DGN-004 / DEF-003
Table 40 - Photographic evidence assessment - PHOTO-DGN-004

Observation	Engineering assessment	Recommended action
A localized polished/worn contact band is visible on one side of the Deflector-Guide Sheave North groove, showing an asymmetric wear/contact distribution within the working groove area. The handwritten 65 mm groove-depth field record is visible on the sheave and agrees with the inspection field notes.	The concentrated sidewall wear is consistent with repeated off-center rope contact and with a possible alignment-related wear mechanism. The 65 mm value provides a traceable as-found groove-depth record, but remaining service margin cannot be established without verified nominal/new groove geometry and OEM/client acceptance criteria.	Retain 65 mm as the as-found field value and repeat/verify the depth at an identified indexed position using recorded measuring equipment. Compare the result and full groove profile with verified OEM/client criteria as part of the corrective intervention.

16.3.13 Photographic Evidence Sheet**Photo ID**

PHOTO-DGN-005

Finding ID

DEF-003

Asset

Deflector-Guide Sheave North

Component / location

Rope groove - marked profile

View / direction

Marked groove reference view

Date / time

18/07/2026 (inspection date)

Caption

Marked groove view illustrating lateral offset of the worn contact profile - North.

*Figure 22 - Deflector-Guide Sheave North evidence - PHOTO-DGN-005 / DEF-003**Table 41 - Photographic evidence assessment - PHOTO-DGN-005*

Observation	Engineering assessment	Recommended action
The marked reference view shows the visible worn/contact band displaced laterally relative to the indicated groove reference/center region.	The lateral displacement provides qualitative evidence of an off-center operating contact pattern. It is consistent with angular rope entry from Drive Wheel North and may also be compatible with a non-zero exit fleet angle on the downstream side, although that broader mechanism is not confirmed by this photograph alone.	Use a formal alignment datum and dimensional groove survey to quantify the offset, review both the entry and exit rope path relationships, correct the sheave relationship, and establish a documented post-correction baseline.

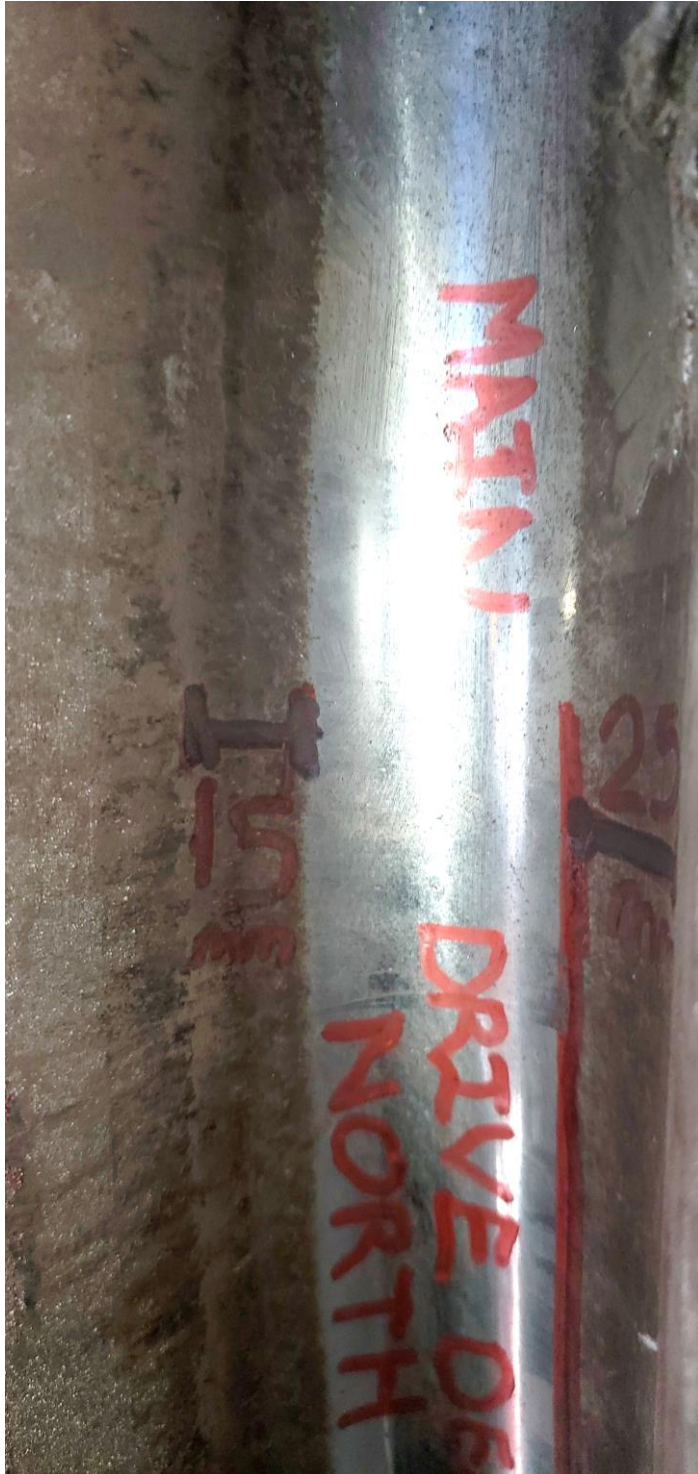
16.3.14 Photographic Evidence Sheet**Photo ID**

PHOTO-DGN-006

Finding ID

DEF-003

Asset

Deflector-Guide Sheave North

Component / location

Rope groove - marked profile

View / direction

Additional marked groove reference view

Date / time

18/07/2026 (inspection date)

Caption

Additional marked view illustrating lateral offset of the worn contact profile - North.

Figure 23 - Deflector-Guide Sheave North evidence - PHOTO-DGN-006 / DEF-003

Table 42 - Photographic evidence assessment - PHOTO-DGN-006

Observation	Engineering assessment	Recommended action
The additional marked view shows that the visible worn/contact region is not centered on the indicated groove reference position.	This view corroborates the lateral profile offset documented in PHOTO-DGN-005 and is consistent with sustained angular rope entry. It also supports the need to verify whether the downstream exit rope path is contributing to the observed groove-contact pattern.	Retain this image as baseline evidence, complete the alignment and groove-profile surveys, assess the downstream exit path, and repeat comparable photographs after corrective work.

16.4 Deflector-Guide Sheave South

Defect Ref ID	DEF-004	Asset	Deflector-Guide Sheave South
Finding title	Off-center groove contact and asymmetric wear consistent with angular rope entry - South	Location	Cable Belt Main Drive - South

16.4.1 Asset Overview and General Condition

Deflector-Guide Sheave South | Cable Belt Main Drive - South | Overall condition: Poor | Priority engineering review required. The condition rating reflects visible off-center groove contact and localized asymmetric deterioration within the rope-contact profile. No critical fracture or catastrophic component failure is established by the supplied photographic evidence; however, the observed contact and wear mechanism requires corrective investigation and verification.

16.4.2 Inspection Observation

PHOTO-DGS-001 and PHOTO-DGS-002 show the groove profile gauge seated within the Deflector-Guide Sheave South groove with visibly unequal contact across the profile. The marked reference features indicate an approximate groove center/reference position, while the visible operating contact/wear region is displaced from that centered region.

PHOTO-DGS-003 documents the groove surface condition. PHOTO-DGS-004 provides a marked groove/profile reference view and shows the worn/contact profile displaced relative to the indicated center/reference position.

Inspection field notes record an as-found groove depth of 66 mm and approximately 2 mm local groove wear on the Deflector-Guide Sheave South. No photograph of the measuring operation was retained.

The evidence provides strong qualitative evidence of an asymmetric and laterally offset groove-contact pattern. The recorded 66 mm groove depth and approximately 2 mm local wear are as-found field measurements. The Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurement; however, the measurement method/instrument used for the approximate 2 mm local-wear value was not recorded. Verified nominal/new groove geometry, remaining wear allowance, external alignment measurements and OEM/client discard limits were not supplied.

16.4.3 Rope Groove and Rim Condition

The Deflector-Guide Sheave South groove exhibits asymmetric wear and an off-center operating contact profile. The profile-gauge photographs and marked reference view indicate that the visible worn/contact region is displaced from the approximate groove center/reference position rather than being distributed symmetrically about the intended groove profile.

Inspection field notes record an as-found groove depth of 66 mm and approximately 2 mm local groove wear. These values quantify the observed condition at the recorded inspection location but do not, by themselves, establish remaining wear allowance or acceptance because verified nominal/new groove geometry and OEM/client discard criteria were not supplied. The full circumferential extent and dimensional profile of the groove also remain to be mapped.

Groove condition: Pronounced asymmetric wear and profile offset | Recorded groove depth: 66 mm | Approximate local wear: ~2 mm | Remaining service margin against OEM/client criteria not established.

16.4.4 Rope Tracking and Alignment

Rope tracking: The profile-gauge and marked groove evidence shows that the operating contact band is displaced from the indicated centered groove/contact region. When considered together with the asymmetric Drive Wheel South groove/liner deterioration, the condition is consistent with the possibility that the rope approaches the Deflector-Guide Sheave South on an angular path from Drive Wheel South rather than following a common centered sheave/rope relationship.

A further possible contributing factor is that the rope may also leave the Deflector-Guide Sheave South with a non-zero downstream fleet angle after the sheave. No downstream guide arrangement or measured exit-path datum was supplied to confirm or exclude this condition. This possible mechanism should therefore be treated as a system-level hypothesis requiring verification by external alignment measurements and rope-path survey.

Alignment limitation: No formal laser, optical or dimensional alignment survey was supplied. The presence of any entry or exit rope-path geometry/alignment error, and if present its magnitude, direction and geometric source, remain to be established by the formal survey.

Overall status: Abnormal groove-contact pattern consistent with possible angular rope entry | Possible non-zero exit fleet angle to be verified | Formal geometry/alignment verification required | Corrective action to be defined from the survey results.

16.4.5 Hub, Shaft and Bearing Housing

The supplied evidence does not establish the internal condition of the Deflector-Guide Sheave South bearings or shaft. No bearing temperature, vibration, axial-position/end-float, clearance or shaft-alignment measurements were supplied.

The observed off-center contact pattern has the potential to introduce non-uniform lateral loading into the sheave/support system; however, no bearing load or resulting bearing/shaft damage was measured or confirmed during this inspection. Bearing and shaft condition should therefore be verified as part of the alignment-correction and follow-up assessment.

16.4.6 Structural Components, Fasteners and Guards

No material structural defect was specifically identified from the supplied evidence associated with this finding. The available photographs are not a substitute for a dedicated structural, fastener or NDT inspection. Accessible supports, mounting fasteners, guards and associated interfaces should be checked during the alignment-correction intervention and any additional defects recorded separately.

16.4.7 Corrosion, Coating and Housekeeping

No separate material corrosion or coating defect was quantified from the supplied evidence for this finding. Surface contamination is present within the groove region and may obscure local wear boundaries. The rope-contact surfaces should be cleaned before detailed dimensional/profile assessment so that the as-found groove condition and material loss can be reliably documented and retained as the pre-replacement baseline.

16.4.8 Engineering Assessment

ENGINEERING INTERPRETATION

The Deflector-Guide Sheave South evidence demonstrates a pronounced off-center groove-contact condition. The profile-gauge photographs show unequal contact within the groove, while the marked reference view shows that the visible worn/contact region is displaced from the approximate groove center/reference position. The observed wear is therefore not consistent with a nominally centered and symmetrical rope-contact pattern.

Inspection field notes record an as-found groove depth of 66 mm and approximately 2 mm local groove wear. No photograph of the measuring operation was retained. These readings provide quantitative as-found condition evidence. The Bosch PLR 50, equipment ID 009741746, is recorded as the instrument used for the groove-depth measurement; however, the measurement method/instrument used for the approximate 2 mm local-wear value was not recorded. Verified nominal/new groove geometry and OEM/client discard criteria were not supplied. The remaining wear allowance therefore cannot be calculated from these values alone.

When considered together with the Drive Wheel South findings, the observed condition is consistent with the possibility that the rope enters the Deflector-Guide Sheave South on an angular path from Drive Wheel South rather than along a common centered sheave/rope relationship. Repeated off-center entry provides a credible mechanism for the asymmetric sidewall contact, polishing and displaced operating profile observed in the South guide-sheave groove.

A further system-level possibility is that the rope may also be leaving the Deflector-Guide Sheave South with a non-zero downstream fleet angle. No measured downstream rope-path datum or external alignment survey was supplied to confirm this condition, and no conclusion should therefore be made from the wear pattern alone. The possible exit-side contribution should be assessed together with the entry relationship during the integrated alignment review of the complete rope path.

The wear pattern can therefore be interpreted in two complementary ways: an incorrect entry and/or exit rope-path relationship could be contributing to the irregular groove wear, while the irregular wear pattern itself provides qualitative evidence that the rope has not been operating centrally within the intended groove profile. The field

measurements quantify the as-found groove condition but do not establish whether geometric misalignment is present or the remaining wear allowance against a verified OEM/client criterion. No conclusion regarding bearing or shaft damage can be made from the available evidence.

On the evidence presently available, the Deflector-Guide Sheave South condition is considered to require priority corrective investigation. A formal alignment survey of the Drive Wheel South / Deflector-Guide Sheave South relationship and the downstream exit rope path should be performed using a documented reference datum. Before removal/replacement, the 66 mm depth and approximately 2 mm local wear should be repeated at identified/indexed locations where practicable using recorded measuring equipment, then compared with verified OEM/client criteria.

The South 45 mm cable-belt rope should also be inspected in the corresponding contact region for localized abrasion, wire damage, distortion or other evidence of sustained side contact. Following correction, the rope path and groove/contact profile should be rechecked and documented as the new baseline.

Condition classification: Poor | Level 3 - Significant | P2 - Urgent.

16.4.9 Defect Classification and Recommended Action

Table 43 - Deflector-Guide Sheave South finding classification and action

Defect ID	Condition rating	Severity	Priority	Recommended action	Responsible discipline	Verification requirement
DEF-004	Poor	Level 3 - Significant	P2 - Urgent	Perform a formal alignment survey of the Drive Wheel South / Deflector-Guide Sheave South pair and the downstream exit rope path using a documented datum, and correct the rope-path/sheave alignment where a non-compliant geometry condition is confirmed. Retain the recorded as-found groove values of 66 mm depth and approximately 2 mm local wear; repeat/verify them at identified indexed locations where practicable using recorded measuring equipment. Compare the dimensional groove profile with verified OEM/client criteria and complete the recommended Deflector-Guide Sheave replacement as part of the coordinated corrective intervention. Inspect the South 45 mm cable-belt rope for localized side-contact damage.	Mechanical / alignment / rope specialist	Recorded alignment survey/correction report; repeatable groove-depth/local-wear measurements with equipment ID and indexed location; comparison with verified OEM/client criteria; replacement identification and final groove/profile baseline; post-correction rope-tracking evidence; rope inspection and close-out record.

16.4.10 Photographic Evidence Sheet



Photo ID
 PHOTO-DGS-001

Finding ID
 DEF-004

Asset
 Deflector-Guide Sheave South

Component / location
 Rope groove / profile

View / direction
 Close-up profile-gauge view

Date / time
 18/07/2026 (inspection date)

Caption
 Profile-gauge view showing the worn
 Deflector-Guide Sheave South groove.

Figure 24 - Deflector-Guide Sheave South evidence - PHOTO-DGS-001 / DEF-004

Table 44 - Photographic evidence assessment - PHOTO-DGS-001

Observation	Engineering assessment	Recommended action
The profile gauge is shown seated within the Deflector-Guide Sheave South groove. The marked reference indicates the approximate groove center/reference position, while the visible gauge contact is unequal across the groove profile.	The unequal gauge contact provides qualitative evidence that the operating groove profile is asymmetric and displaced from a nominal centered contact condition. The photograph does not establish the magnitude of wear, the gauge geometry or the remaining service margin.	Complete the South sheave-pair alignment survey and dimensional groove-profile assessment using identified measuring equipment and verified acceptance criteria. Retain this image as baseline evidence for post-correction comparison.

16.4.11 Photographic Evidence Sheet



Photo ID
 PHOTO-DGS-002

Finding ID
 DEF-004

Asset
 Deflector-Guide Sheave South

Component / location
 Rope groove / profile

View / direction
 Additional close-up profile-gauge view

Date / time
 18/07/2026 (inspection date)

Caption
 Additional profile-gauge view of the Deflector-Guide Sheave South groove.

Figure 25 - Deflector-Guide Sheave South evidence - PHOTO-DGS-002 / DEF-004
Table 45 - Photographic evidence assessment - PHOTO-DGS-002

Observation	Engineering assessment	Recommended action
The additional profile-gauge view shows the South groove from an alternative position and again demonstrates visibly unequal contact across the groove profile.	This photograph corroborates PHOTO-DGS-001 and supports the conclusion that the groove-contact profile is asymmetric rather than centered and uniform. Quantitative groove geometry and verified OEM/client criteria remain required to characterize the as-found condition and support the pre-replacement documentation and final close-out.	Record dimensional profile measurements after alignment verification/correction and retain the results as the as-found baseline before the recommended Deflector-Guide Sheave South replacement.

16.4.12 Photographic Evidence Sheet



Photo ID
 PHOTO-DGS-004

Finding ID
 DEF-004

Asset
 Deflector-Guide Sheave South

Component / location
 Rope groove - marked profile

View / direction
 Marked groove profile reference view

Date / time
 18/07/2026 (inspection date)

Caption
 Marked view illustrating the offset/asymmetric groove contact profile - South.

Figure 26 - Deflector-Guide Sheave South evidence - PHOTO-DGS-004 / DEF-004
Table 46 - Photographic evidence assessment - PHOTO-DGS-004

Observation	Engineering assessment	Recommended action
The marked South groove view identifies the approximate groove center/reference position and shows the visible worn/contact profile displaced laterally from that indicated centered region.	The lateral profile displacement provides qualitative evidence of sustained off-center rope contact and is consistent with angular rope entry from Drive Wheel South. In the broader system context, it also supports the need to verify whether the downstream exit rope path is contributing to the contact pattern; that possible mechanism is not confirmed by this photograph alone.	Use a formal alignment datum and dimensional groove survey to quantify the offset, review both the entry and downstream exit rope-path relationships, correct the sheave/rope-path geometry as required, and retain a post-correction photograph/profile record as the new maintenance baseline.

17 Summary of Defects

The four recorded defects are retained as separate asset findings for traceability, but the combined evidence supports treating them as potentially mechanically related manifestations of a broader rope-path geometry condition. The North and South wear patterns should therefore be evaluated as one integrated drive-system geometry problem, with the presence and magnitude of any misalignment or fleet-angle error to be established by survey before corrective repositioning is finalized. For the Deflector-Guide Sheaves, field notes additionally record as-found groove depth/local-wear values of 65 mm / approximately 3 mm (North) and 66 mm / approximately 2 mm (South). These measurements quantify the local as-found condition but do not establish acceptance or remaining wear allowance without verified OEM/client groove criteria.

Table 47 - Consolidated defect register

Defect ID	Asset	Component / location	Defect / observation	Condition	Severity	Priority	Evidence	Responsible discipline	Status
DEF-001	Drive Wheel North	Rope groove / liner and rope-path alignment	Off-center rope departure with severe asymmetric liner deterioration; condition is consistent with non-uniform rope loading and requires verification of both incoming and inter-sheave fleet angles.	Poor	Level 3	P2	PHOTO-DWN-001 to 004; CBDS-REF002 to 004; CBDS-REF009/010	Mechanical / alignment / rope specialist	Open
DEF-002	Drive Wheel South	Rope groove / liner and rope-path alignment	Pronounced asymmetric liner deterioration, mirror-imaged relative to North; condition is consistent with non-uniform rope loading and possible entry/inter-sheave fleet-angle error.	Poor	Level 3	P2	PHOTO-DWS-001/002; CBDS-REF002 to 004; CBDS-REF009/010	Mechanical / alignment / rope specialist	Open
DEF-003	Deflector-Guide Sheave North	Rope groove and entry/exit rope-path alignment	Pronounced asymmetric groove/profile displacement with recorded 65 mm groove depth and approximately 3 mm local wear. Condition is consistent with possible angular entry and possible downstream exit fleet-angle contribution. South32 condition monitoring reportedly also identified an indication consistent with misalignment; supporting data require review.	Poor	Level 3	P2	PHOTO-DGN-002 to 006; inspection field notes; CBDS-REF002 to 004; CBDS-REF011 to 013	Mechanical / alignment / condition monitoring	Open

Defect ID	Asset	Component / location	Defect / observation	Condition	Severity	Priority	Evidence	Responsible discipline	Status
DEF-004	Deflector-Guide Sheave South	Rope groove and entry/exit rope-path alignment	Pronounced asymmetric/off-center groove profile with recorded 66 mm groove depth and approximately 2 mm local wear. Condition is consistent with possible angular entry; downstream exit fleet angle remains to be quantified.	Poor	Level 3	P2	PHOTO-DGS-001/002/004; inspection field notes; CBDS-REF002 to 004; CBDS-REF011 to 013	Mechanical / alignment / rope specialist	Open

18 Inspection Observation Log

Table 48 - Inspection observation and evidence log

Observation ID	Asset	Location	Observation	Evidence reference	Condition rating	Related defect ID	Status
OBS-001	Drive Wheel North	Rope exit	Rope path is visibly off-center at the sheave exit.	PHOTO-DWN-001 / 002	Poor	DEF-001	Open
OBS-002	Drive Wheel North	Groove / liner	Localized sidewall/liner damage is visible near the exit region.	PHOTO-DWN-003 / 004	Poor	DEF-001	Open
OBS-003	Drive Wheel South	Groove / liner	Left/right wear is asymmetric, with one side showing more pronounced deterioration.	PHOTO-DWS-001 / 002	Poor	DEF-002	Open
OBS-004	Deflector-Guide Sheave North	Rope groove	Profile-gauge and groove-surface views show asymmetric profile wear. Inspection field notes record 65 mm groove depth and approximately 3 mm local wear.	PHOTO-DGN-002 / 003 / 004; inspection field notes	Poor	DEF-003	Open
OBS-005	Deflector-Guide Sheave North	Rope groove	Marked reference views show lateral offset of the worn contact profile.	PHOTO-DGN-005 / 006 / 007	Poor	DEF-003	Open
OBS-006	Deflector-Guide Sheave South	Rope groove	Profile-gauge views and marked groove reference show asymmetric / offset wear. Inspection field notes record 66 mm groove depth and approximately 2 mm local wear.	PHOTO-DGS-001 / 002 / 004; inspection field notes	Poor	DEF-004	Open

Observation ID	Asset	Location	Observation	Evidence reference	Condition rating	Related defect ID	Status
OBS-007	Cable Belt Upper Incoming Guide Sheaves	North and South incoming rope paths	CBIGS photographs identify the guide-sheave assemblies that control the rope approach into the Drive Wheels. No fleet-angle measurement is established by the photographs alone.	CBIGS-001 / 002 / 003; CBDS-REF009/010	Information / context	DEF-001 / DEF-002	Verification required
OBS-008	Deflector-Guide Sheave North	Condition-monitoring context	South32 Wessels Mine Condition Monitoring Supervisor reported vibration-analysis evidence indicating apparent misalignment at Deflector-Guide Sheave North. The underlying trace/report was not supplied for independent review.	Reported operational context; data outstanding	Information / supporting context	DEF-003	Supporting data outstanding

19 Engineering Assessment

CONSOLIDATED ENGINEERING ASSESSMENT

System-level interpretation. The four asset findings form a coherent system-level pattern. Drive Wheel North shows visible off-center rope departure and asymmetric liner damage; Drive Wheel South shows the corresponding mirror-image asymmetric liner condition; and both Deflector-Guide Sheaves exhibit laterally displaced, non-symmetrical groove-contact profiles. Inspection field notes also quantify the as-found Deflector-Guide groove condition at selected locations: North 65 mm groove depth with approximately 3 mm local wear, and South 66 mm groove depth with approximately 2 mm local wear. These conditions are not characteristic of a rope remaining consistently centered through the full drive arrangement. The most credible consolidated hypothesis is that one or more non-zero fleet angles within the rope path may be contributing to repeated side loading and preferential groove/liner contact. This hypothesis requires quantitative verification, and the present evidence does not establish which, if any, interface is outside the verified tolerance.

Geometry to be quantified. The formal rope-path geometry remains unmeasured even though selected Deflector-Guide groove condition readings are now recorded. The present photographic/field-note evidence does not identify which interface is the primary geometric cause or the relative contribution of each interface. A quantitative survey is therefore required across six locations: (1) entry into Drive Wheel North from the Cable Belt Upper Incoming Guide Sheave North; (2) entry into Drive Wheel South from the Cable Belt Upper Incoming Guide Sheave South; (3) Drive Wheel North exit to Deflector-Guide Sheave North entry; (4) Drive Wheel South exit to Deflector-Guide Sheave South entry; (5) Deflector-Guide Sheave North downstream exit; and (6) Deflector-Guide Sheave South downstream exit. The target is a fleet angle of 0 degrees, or as close to 0 degrees as practicable within verified OEM/client tolerances. A common recorded datum should be used; the Drive Wheel groove/shaft centerline may be used as the primary static reference unless verified design information requires a different datum.

Relationship to the reference drawings and physical guide sheaves. CBDS-REF009 and CBDS-REF010 illustrate the desired and possible non-zero Drive Wheel entry fleet-angle condition, while CBIGS-001 to CBIGS-003 identify the actual Upper Incoming Guide Sheaves that influence that approach path. CBDS-REF002, CBDS-REF003 and CBDS-REF004 illustrate the Drive Wheel-to-Deflector relationship. CBDS-REF011 and CBDS-REF012 illustrate the desired and possible non-zero Deflector-Guide exit condition, and CBDS-REF013 shows a conceptual lower overhead guide-sheave arrangement that could be considered if the downstream path cannot otherwise be controlled sufficiently close to zero.

Condition-monitoring correlation. The South32 Wessels Mine Condition Monitoring Supervisor reportedly identified a vibration-analysis indication consistent with misalignment at Deflector-Guide Sheave North. This information is directionally consistent with the visible North-side rope-path and groove-wear evidence. The

underlying vibration data were not supplied, so the report does not independently validate the vibration diagnosis or quantify the misalignment; the records should be obtained and correlated with the geometric survey.

Geometry correction and component renewal should be coordinated. Where the survey confirms an incorrect rope-path geometry, geometry correction and renewal of the worn rope-contact components are interdependent. Replacing sheaves or liners while leaving an incorrect rope path will expose the new components to the same asymmetric loading and is likely to reproduce the accelerated wear mechanism. Conversely, correcting the geometry while retaining severely distorted groove/liner profiles may continue to provide inadequate or non-uniform support to the 45 mm ropes. The corrective intervention should therefore first establish the actual geometry, then correct the relevant sheave/guide positions, and then renew the worn rope-contact components as a coordinated work package.

Recommended component disposition. The visibly damaged liner assemblies on both Drive Wheels should be renewed, with the exact segment/full-liner replacement scope confirmed against the installed liner arrangement and applicable OEM/client requirements. Replacement of both Deflector-Guide Sheaves remains recommended because the existing asymmetric groove profiles have developed a displaced operating contact geometry; retaining those grooves after alignment correction may continue to support the rope incorrectly. The recorded North 65 mm / approximately 3 mm wear and South 66 mm / approximately 2 mm wear values should be retained as baseline evidence of the as-found condition and repeated at indexed locations before removal where practicable. Because verified nominal/new groove geometry and discard limits are not available, these readings are not used as stand-alone acceptance/discard criteria.

Engineering objective and expected outcome. The remedial objective is to restore a controlled rope path through the complete drive unit so the rope is centered within each groove and the relevant fleet angles are 0 degrees, or as close to zero as practicable. Achieving this geometry with renewed rope-contact surfaces should promote substantially more symmetrical groove wear, reduce localized rope abrasion and side loading, extend the service life of the cable belt ropes and sheave components, and provide a more stable condition-monitoring baseline for long-term maintenance planning.

20 Recommendations

REC-001, the quantitative six-interface rope-path geometry/alignment survey, remains the principal follow-up recommendation generated by this inspection report. The corrective strategy is structured as a five-phase sequence so that measurement governs the geometry-dependent work, while the already-recommended renewal of worn rope-contact components is incorporated into one coordinated intervention. The system-level recommendations intentionally link geometry correction with component renewal: replacing worn surfaces without correcting any confirmed rope-path geometry error may reproduce the wear mechanism, while geometry correction alone will not restore suitable rope support where the existing groove/liner contact profiles have already been permanently distorted.

20.1 Corrective Objective

The corrective objective is to restore a controlled rope path through the complete Cable Belt Main Drive arrangement so that the rope is centered within the relevant sheave groove and the fleet angle at each controlled interface is 0° or as close to 0° as practicable within verified OEM/client tolerances. The corrective strategy must therefore first quantify the actual geometry, then define and validate the required configuration, execute the approved work as one coordinated intervention, and finally verify the completed installation and establish a new condition baseline.

Core principle

The Phase 1 survey is the engineering decision gate. It determines what must be repositioned, by how much, and whether additional rope-control components are required. It does not determine whether the already-recommended rope-contact component renewals are required; those renewals form part of the coordinated corrective scope once the intervention proceeds.

20.2 Recommendation Logic

Stage	Status	Function
Phase 1	MANDATORY - decision gate	Measure and quantify all six rope-path interfaces from a common recorded datum.
Phase 2	DYNAMIC / survey-dependent	Convert Phase 1 results into the exact corrective geometry: what moves, what does not move, and the required final positions.
Phase 3	CONDITIONAL engineering / mandatory planning close-out	Engineer and validate any additional modifications required by the Phase 1/2 results and prepare the complete shutdown work package.
Phase 4	MANDATORY coordinated intervention	Execute all approved geometry corrections plus renewal of both Drive Wheel liner assemblies and replacement of both Deflector-Guide Sheaves.
Phase 5	MANDATORY acceptance / baseline	Remeasure, verify the completed rope path, and establish the new zero-time condition baseline before corrective close-out.

20.3 Five-Phase Corrective Sequence

20.3.1 Phase 1 - Quantify the Complete Rope Path

Perform the quantitative six-interface rope-path geometry/alignment survey using one common, recorded datum. The survey shall establish the actual as-found geometry before final repositioning or modification work is defined.

- Interface 1: Cable Belt Upper Incoming Guide Sheave North to Drive Wheel North rope-entry fleet angle.
- Interface 2: Cable Belt Upper Incoming Guide Sheave South to Drive Wheel South rope-entry fleet angle.
- Interface 3: Drive Wheel North to Deflector-Guide Sheave North centerline/fleet-angle relationship.
- Interface 4: Drive Wheel South to Deflector-Guide Sheave South centerline/fleet-angle relationship.
- Interface 5: Deflector-Guide Sheave North downstream rope-exit fleet angle.
- Interface 6: Deflector-Guide Sheave South downstream rope-exit fleet angle.
- Use the same recorded datum and reference convention for the pre-correction and final verification surveys.
- Record the applicable verified OEM/client tolerances used to judge each interface.
- Do not determine final repositioning distances from photographs or schematic reference drawings alone.

Phase 1 required output

A complete numerical geometry record identifying each interface as acceptable / correction required, together with the quantified direction and amount of any required repositioning and the measured basis for any additional rope-control modification.

20.3.2 Phase 2 - Define the Corrective Configuration from the Survey Results

Use the Phase 1 measurements to convert the observed condition into a defined corrective geometry. Phase 2 is intentionally dynamic: an interface that is already within the verified tolerance requires no repositioning, while an interface outside tolerance is assigned a quantified correction.

- Reposition Cable Belt Upper Incoming Guide Sheave North and/or South only where the corresponding Drive Wheel entry fleet angle is outside the verified tolerance.
- Determine the final installed position of Deflector-Guide Sheave North and South from the measured Drive Wheel-to-Deflector relationship. The replacement sheaves must not simply reproduce the existing mounting geometry unless that geometry is verified as acceptable.
- Quantify the downstream exit condition on both sides and determine whether the existing arrangement can achieve the required exit fleet-angle target without additional guide components.
- Record "no change required" where a surveyed interface is already acceptable, so the work scope is based on measured need rather than assumption.

Phase 2 required output

An approved corrective geometry schedule showing the final required position/configuration for each affected component and clearly identifying which potential changes are not required.

20.3.3 Phase 3 - Engineer, Validate and Prepare the Complete Corrective Work Package

Where the Phase 1/2 results identify requirements beyond simple adjustment, complete the necessary engineering design and planning before the corrective shutdown is executed.

- If either Deflector-Guide exit path cannot be brought sufficiently close to 0° using the existing arrangement, perform a formal design review for an engineered rope-control modification. CBDS-REF013 remains a conceptual option only; any lower overhead guide-sheave installation requires approved geometry, structural support, guarding, access and maintainability review.
- Confirm the replacement specification and installation requirements for Deflector-Guide Sheave North and South, including incoming groove geometry and the survey-defined final installation position.
- Confirm the Drive Wheel North and South liner-renewal scope against the installed liner arrangement and applicable OEM/client requirements.
- Prepare procurement, access, lifting, alignment, isolation, guarding, inspection and commissioning requirements so that the complete corrective scope can be executed as one coordinated intervention.
- Obtain/review available condition-monitoring and vibration records, particularly the reported Deflector-Guide Sheave North indication, so that the pre-correction condition is available for comparison.

Phase 3 required output

A fully defined and approved shutdown work package with no unresolved geometry/design question that could reasonably be closed before the intervention.

20.3.4 Phase 4 - Execute the Coordinated Corrective Shutdown

Execute the approved geometry corrections, component renewals and associated inspections as one integrated intervention. The geometry-dependent work is performed only where required by Phases 1-3; the rope-contact component renewal listed below is treated as mandatory within the recommended corrective work package.

- Reposition the Cable Belt Upper Incoming Guide Sheaves where the Phase 1/2 survey identifies a required entry-path correction.
- Install the new Deflector-Guide Sheaves at the final survey-defined positions required to achieve the intended Drive Wheel-to-Deflector relationship.
- Install any approved lower overhead guide sheaves or other engineered exit-path control modification where required by the Phase 1-3 outcome.
- Renew the Drive Wheel North liner assembly.
- Renew the Drive Wheel South liner assembly.
- Replace Deflector-Guide Sheave North.
- Replace Deflector-Guide Sheave South.
- Inspect both nominal 45 mm ropes for side-loading damage, abrasion, broken wires, strand deformation, crushing/pinching and diameter/profile change.
- Verify bearing/shaft/support condition and capture relevant pre/post condition-monitoring data during the intervention.
- Where practicable before removal, repeat the as-found Deflector-Guide groove readings at identified indexed positions and record equipment identification for traceability.

Deflector-Guide replacement basis

North replacement is condition-driven by the pronounced asymmetric/off-center groove condition, the recorded 65 mm groove depth / approximately 3 mm local wear, and the reported condition-monitoring concern. South shows the same established asymmetric/off-center operating profile but with less severe recorded local wear (66 mm groove depth / approximately 2 mm). The available evidence does not establish that the South sheave has exhausted all residual life against a verified discard limit; however, replacing it during the same coordinated shutdown avoids retaining a pre-worn rope-contact geometry in the corrected system, reduces the likelihood of a subsequent dedicated replacement shutdown, and establishes a common renewed baseline across both Deflector-Guide Sheaves.

20.3.5 Phase 5 - Verify the Corrected System and Establish the New Baseline

The engineering objective is not considered demonstrated merely because the components have been replaced. The completed installation must be quantitatively verified before corrective close-out and the post-intervention condition must be recorded as the new maintenance baseline.

- Remeasure all six rope-path interfaces using the same datum/reference convention used in Phase 1.
- Confirm centered rope tracking and fleet angles of 0° or as close to 0° as practicable within the verified applicable tolerances.
- Record the final as-installed positions of the Upper Incoming Guide Sheaves, Deflector-Guide Sheaves and any added lower overhead guide sheaves.
- Capture the renewed Drive Wheel liner condition/profile and incoming/new Deflector-Guide groove geometry.
- Retain post-correction rope-tracking photographs and rope-condition records.
- Retain post-correction bearing/shaft condition-monitoring and vibration records where available.
- Compile the final alignment, installation, inspection and commissioning information into a controlled baseline pack for future wear trending and predictive maintenance.

Phase 5 close-out criterion

Corrective close-out is achieved when the approved configuration has been installed, the six-interface verification confirms the intended rope-path geometry within the verified tolerances, and the complete post-correction baseline has been recorded for future trending.

20.4 Detailed Recommendation and Action Register

The existing REC-001 to REC-009 structure is retained. The five-phase logic above governs how each recommendation is applied: Phase 1 is the mandatory measurement gate; geometry changes in Phases 2-3 are survey-dependent; component renewal in Phase 4 is mandatory within the coordinated corrective work package; and Phase 5 is the mandatory verification and baseline close-out gate.

Table 49 - Detailed recommendation and action register

ID	Phase	Recommendation	Applicability	Verification / close-out
REC-001	1	Survey and record all six entry, inter-sheave and exit interfaces using one common datum. Record the applicable verified tolerances and quantify the direction and amount of any required repositioning.	Mandatory decision gate	Pre-correction six-interface geometry/alignment report.
REC-002	2	Reposition the North/South Cable Belt Upper Incoming Guide Sheaves only where the measured Drive Wheel entry fleet angle is outside the verified tolerance. Record no change required where an interface is already acceptable.	Conditional - Phase 1 outcome	Before/after entry-angle readings and adjustment record.
REC-003	2	Define and achieve the required Drive Wheel-to-Deflector relationship. The final position of each replacement Deflector-Guide Sheave is to be based on the Phase 1 survey, not assumed from the existing installation.	Conditional positioning / mandatory final setting	Before/after centerline and fleet-angle readings; installation/alignment record.
REC-004	2-3	Where the downstream Deflector-Guide exit path cannot achieve the verified target with the existing arrangement, complete a formal design review and install an approved rope-control modification such as lower overhead guide sheaves if required. CBDS-REF013 remains conceptual only.	Conditional - Phase 1/2 outcome	Exit-angle readings; approved design/change record where applicable; final angle verification.
REC-005	4	Renew the Drive Wheel North and South liner assemblies during the coordinated corrective shutdown; confirm the exact installed liner-set/segment scope against the applicable arrangement and OEM/client requirements.	Mandatory within corrective work package	Replacement scope, installation record and renewed liner/profile baseline.
REC-006	4	Replace Deflector-Guide Sheave North and South during the coordinated corrective shutdown. Repeat/verify as-found groove readings at indexed positions before removal where practicable, verify incoming groove geometry, and install each sheave in its survey-defined final position.	Mandatory within corrective work package	Pre-removal/as-found record where practicable; replacement ID; new groove/profile and final alignment record.
REC-007	4	Inspect both nominal 45 mm ropes for evidence of side-loading damage and determine disposition against approved criteria.	Mandatory inspection during intervention	Rope condition/disposition report.
REC-008	3-5	Obtain/review available vibration data and verify bearing/shaft/support condition before and after correction, particularly the reported Deflector-Guide Sheave North indication.	Mandatory verification where records/access permit	Pre/post condition-monitoring and mechanical checks; trend/close-out record.
REC-009	5	Remeasure all six interfaces using the Phase 1 datum, confirm centered rope tracking within verified tolerances, record the final installed configuration, and establish the final photographic/profile/rope/condition-monitoring baseline.	Mandatory close-out gate	Final six-interface verification and commissioning/baseline pack.

20.5 Recommended Sequence - One-Line Summary

MEASURE → DEFINE → ENGINEER / PREPARE → EXECUTE → VERIFY & BASELINE

In practical terms: Phase 1 quantifies the need; Phases 2-3 define and validate the final configuration; Phase 4 executes the required geometry work together with mandatory component renewal; and Phase 5 verifies the result and establishes the new zero-time baseline.

Success state

A corrected and renewed rope-contact system with verified geometry, centered rope tracking, fleet angles at 0° or as close to 0° as practicable within verified tolerances, and a controlled baseline for future wear and condition trending.

21 Conclusion

The inspection findings, considered together, are consistent with a common rope-path geometry condition rather than only unrelated local wear; however, the specific geometry and any misalignment or fleet-angle error have not yet been quantified. The Drive Wheels exhibit asymmetric liner deterioration, while the Deflector-Guide Sheaves exhibit pronounced off-center groove wear. Inspection field notes additionally record the Deflector-Guide Sheave North groove at 65 mm depth with approximately 3 mm local wear and the South groove at 66 mm depth with approximately 2 mm local wear. These readings quantify the as-found local groove condition but do not establish remaining wear allowance or acceptance because verified nominal/new groove geometry and OEM/client discard criteria were not supplied. The overall wear pattern is qualitatively consistent with sustained lateral rope loading that may arise from one or more non-zero fleet angles through the system. The reported South32 condition-monitoring indication of apparent Deflector-Guide Sheave North misalignment is consistent with this interpretation, but the underlying vibration record and the geometric alignment values remain outstanding and must be verified.

The principal corrective objective is to restore the rope path so that the groove centerline and rope travel are coincident through the relevant entry, inter-sheave and exit interfaces, achieving a fleet angle of 0 degrees or as close to 0 degrees as practicable within verified tolerances. This requires a quantitative six-interface alignment/fleet-angle survey before final corrective repositioning work is defined. The survey remains an open follow-up recommendation generated by this report and is not a prerequisite to formal issue of the inspection report. Depending on the measured results, corrective work may include repositioning the Cable Belt Upper Incoming Guide Sheaves, repositioning the Deflector-Guide Sheaves relative to the Drive Wheels, and, where downstream exit control remains inadequate, an engineered modification such as lower overhead guide sheaves following formal design review.

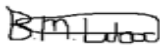
The alignment correction and component-renewal work should be treated as one coordinated intervention. The damaged Drive Wheel liner assemblies on both North and South sides should be renewed, and replacement of both Deflector-Guide Sheaves is recommended so that the corrected system is commissioned with geometrically suitable rope-contact profiles. Replacing worn components without correcting any confirmed rope-path geometry error could reproduce the same asymmetric wear mechanism, while correcting the geometry but retaining the existing distorted grooves may continue to provide non-uniform rope support. Both 45 mm ropes and the associated bearing/shaft systems should also be verified, with available condition-monitoring data correlated to the final alignment results.

Following completion, all six rope-path interfaces should be remeasured and documented, and post-correction rope-tracking, groove/liner profile and condition-monitoring records should be retained as the new maintenance baseline. The intended outcome is more symmetrical sheave wear, reduced rope abrasion and side loading, improved rope and sheave service life, and a more stable basis for predictive maintenance and long-term operating-cost control. The four inspected sheaves remain classified as Poor, Level 3 - Significant, P2 - Urgent pending corrective action and verification. This report does not declare the system fit for continued operation; the operational decision remains with the South32 responsible-site authority.

22 Technical Review and Report Issue Sign-off

This report has completed the internal technical review, document-control confirmation and evidence-traceability review required for controlled issue and is approved within the stated scope and limitations. The quantitative alignment/fleet-angle survey and the other measurement, OEM-criteria, rope-condition and bearing/shaft verification actions identified in this report are post-issue follow-up recommendations arising from the inspection and are not prerequisites to report issue. They remain open actions to be completed through the subsequent corrective-action and finding close-out process. Approval and controlled issue of this report confirm the technical adequacy of the documented assessment and recommendations within the stated scope and limitations; they do not certify completion of the recommended corrective work or constitute a declaration of fitness for continued operation.

Table 50 - Technical review and issue sign-off

Role	Name	Title / discipline	Review status	Date	Signature
Prepared by	Brendan Wood	Projects / Systems	Report preparation	02/08/2026	
Technical reviewer	Kevin Wood	Managing Engineer	Review Complete	11/08/2026	
Approved for issue	Kevin Wood	Managing Engineer	Approval	12/08/2026	

SECTION D

APPENDICES &
EVIDENCE

Photographic records, checklists, measurements, NDT,
drawings and supporting documentation.



Appendix A - Photographic Register

The register below lists the inspection and contextual photographs included in this report. Source filenames are retained for traceability even where a report Photo ID has been standardized. The CBIGS series is included as contextual reference imagery to identify the Cable Belt Upper Incoming Guide Sheaves for later system-level rope-entry fleet-angle assessment; these photographs do not, by themselves, establish a measured alignment or fleet-angle deviation.

Table 51 - Photographic register

Photo ID	Asset	Component / location	View direction	Description / observation	Related defect ID	Date / time	File name / source	Report location
PHOTO-DWN-001	Drive Wheel North	Rope path / groove / contextual area	Inspection view	Left-side view showing the rope exiting the Drive Wheel North off the groove centerline.	DEF-001	18/07/2026; time not recorded	DWNS-001-Rope Exit Off-Set Left Hand Side.jpg	Section 16 if selected / Appendix A
PHOTO-DWN-002	Drive Wheel North	Rope path / groove / contextual area	Inspection view	Opposing view confirming lateral offset of the rope path at the Drive Wheel North exit.	DEF-001	18/07/2026; time not recorded	DWNS-002-Rope Exit Off-Set Right Hand Side.jpg	Section 16 if selected / Appendix A
PHOTO-DWN-003	Drive Wheel North	Rope path / groove / contextual area	Inspection view	Localized liner/groove damage on the right-hand side of Drive Wheel North.	DEF-001	18/07/2026; time not recorded	DWNS-003-Groove Damage Right Hand Side.jpg	Section 16 if selected / Appendix A
PHOTO-DWN-004	Drive Wheel North	Rope path / groove / contextual area	Inspection view	Localized liner/groove damage on the left-hand side of Drive Wheel North.	DEF-001	18/07/2026; time not recorded	DWNS-004-Groove Damage Left Hand Side.jpg	Section 16 if selected / Appendix A
PHOTO-DWS-001	Drive Wheel South	Rope path / groove / contextual area	Inspection view	Right-hand side of Drive Wheel South showing comparatively lower visible wear.	DEF-002	18/07/2026; time not recorded	DWNS-002-Groove Wear Right Hand Side.jpg	Section 16 if selected / Appendix A
PHOTO-DWS-002	Drive Wheel South	Rope path / groove / contextual area	Inspection view	Left-hand side of Drive Wheel South showing the more pronounced asymmetric wear/contact region.	DEF-002	18/07/2026; time not recorded	DWSS-001-Groove Wear Left Hand Side.jpg	Section 16 if selected / Appendix A

Photo ID	Asset	Component / location	View direction	Description / observation	Related defect ID	Date / time	File name / source	Report location
PHOTO-DWS-003	Drive Wheel South	Rope path / groove / contextual area	Inspection view	Context view of the rear of Drive Wheel South and associated brake assemblies.	Information only	18/07/2026; time not recorded	DWSS-003-Breaks Context Photo.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-001	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Location reference view of Deflector-Guide Sheave North.	Information only	18/07/2026; time not recorded	DGSN-001-Location Reference.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-002	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Profile-gauge view showing the worn Deflector-Guide Sheave North groove; inspection annotation/field notes record approximately 3 mm local groove wear at the indicated location.	DEF-003	18/07/2026; time not recorded	DGSN-002-Worn Groove Profile.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-003	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Additional profile-gauge view of the worn Deflector-Guide Sheave North groove.	DEF-003	18/07/2026; time not recorded	DGSN-003-Worn Groove Profile.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-004	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Groove surface detail showing asymmetric polishing/wear on Deflector-Guide Sheave North; handwritten 65 mm groove-depth field record visible on the sheave.	DEF-003	18/07/2026; time not recorded	DGSN-004-Groove Profile Details.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-005	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Marked groove view illustrating lateral offset of the worn contact profile - North.	DEF-003	18/07/2026; time not recorded	DGSN-005-Groove Off-Set Profile-1.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-006	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Additional marked view illustrating lateral offset of the worn contact profile - North.	DEF-003	18/07/2026; time not recorded	DGSN-006-Groove Off-Set Profile-2.jpg	Section 16 if selected / Appendix A
PHOTO-DGN-007	Deflect or-Guide Sheave North	Rope path / groove / contextual area	Inspection view	Contextual side view of the offset groove profile - North.	DEF-003	18/07/2026; time not recorded	DGSN-007-Groove Off-Set Profile-3.jpg	Section 16 if selected / Appendix A
PHOTO-DGS-001	Deflect or-Guide Sheave South	Rope path / groove / contextual area	Inspection view	Profile-gauge view showing the worn Deflector-Guide Sheave South groove.	DEF-004	18/07/2026; time not recorded	DGSS-001-Worn Groove Profile.jpg	Section 16 if selected / Appendix A
PHOTO-DGS-002	Deflect or-Guide Sheave South	Rope path / groove / contextual area	Inspection view	Additional profile-gauge view of the Deflector-Guide Sheave South groove.	DEF-004	18/07/2026; time not recorded	DGSS-002-Worn Groove Profile.jpg	Section 16 if selected / Appendix A

Photo ID	Asset	Component / location	View direction	Description / observation	Related defect ID	Date / time	File name / source	Report location
PHOTO-DGS-003	Deflect or-Guide Sheave South	Rope path / groove / contextual area	Inspection view	Groove surface detail showing the wear profile on Deflector-Guide Sheave South.	Information only	18/07/2026; time not recorded	DGSS-003-Groove Profile Details.jpg	Section 16 if selected / Appendix A
PHOTO-DGS-004	Deflect or-Guide Sheave South	Rope path / groove / contextual area	Inspection view	Marked view illustrating the offset/asymmetric groove contact profile - South.	DEF-004	18/07/2026; time not recorded	DGSS-004-Groove Profile Off-Set Details.jpg	Section 16 if selected / Appendix A
CBIGS-001	Cable Belt Upper Incoming Guide Sheaves	North / South upper incoming guide-sheave arrangement	Underside contextual view	Contextual view showing the North and South upper incoming guide sheaves and associated cable-belt rope paths.	Information / context only	18/07/2026; time not recorded	CBIGS-001-Cable Belt Upper Incoming Guide Sheave Context.jpg	Appendix A - Cable Belt Upper Incoming Guide Sheaves
CBIGS-002	Cable Belt Upper Incoming Guide Sheave North	Upper incoming guide sheave - North	North-side contextual view	Contextual view of the North upper incoming guide sheave and associated incoming cable-belt rope path.	Information / context only	18/07/2026; time not recorded	CBIGS-002-Cable Belt Upper Incoming Guide Sheave North.jpg	Appendix A - Cable Belt Upper Incoming Guide Sheaves
CBIGS-003	Cable Belt Upper Incoming Guide Sheave South	Upper incoming guide sheave - South	South-side contextual view	Contextual view of the South upper incoming guide sheave and associated incoming cable-belt rope path.	Information / context only	18/07/2026; time not recorded	CBIGS-003-Cable Belt Upper Incoming Guide Sheave South.jpg	Appendix A - Cable Belt Upper Incoming Guide Sheaves

Drive Wheel North



PHOTO-DWN-001

Left-side view showing the rope exiting the Drive Wheel North off the groove centerline.

Related defect: DEF-001



PHOTO-DWN-002

Opposing view confirming lateral offset of the rope path at the Drive Wheel North exit.

Related defect: DEF-001

Figure 27 - Drive Wheel North photographic evidence - PHOTO-DWN-001 / PHOTO-DWN-002



PHOTO-DWN-003
Localized liner/groove damage on the right-hand side of Drive Wheel North.
Related defect: DEF-001

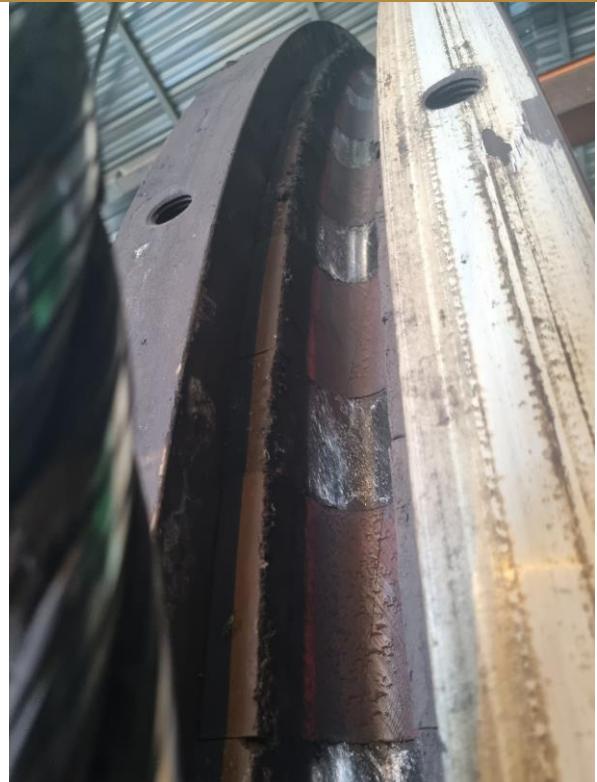


PHOTO-DWN-004
Localized liner/groove damage on the left-hand side of Drive Wheel North.
Related defect: DEF-001

Figure 28 - Drive Wheel North photographic evidence - PHOTO-DWN-003 / PHOTO-DWN-004

Drive Wheel South



PHOTO-DWS-001
Right-hand side of Drive Wheel South showing comparatively lower visible wear.
Related defect: DEF-002



PHOTO-DWS-002
Left-hand side of Drive Wheel South showing the more pronounced asymmetric wear/contact region.
Related defect: DEF-002

Figure 29 - Drive Wheel South photographic evidence - PHOTO-DWS-001 / PHOTO-DWS-002



PHOTO-DWS-003

Context view of the rear of Drive Wheel South and associated brake assemblies.

Related defect: Information / context only

Figure 30 - Drive Wheel South photographic evidence - PHOTO-DWS-003

Deflector-Guide Sheave North



PHOTO-DGN-001
Location reference view of Deflector-Guide Sheave North.
Related defect: Information / context only

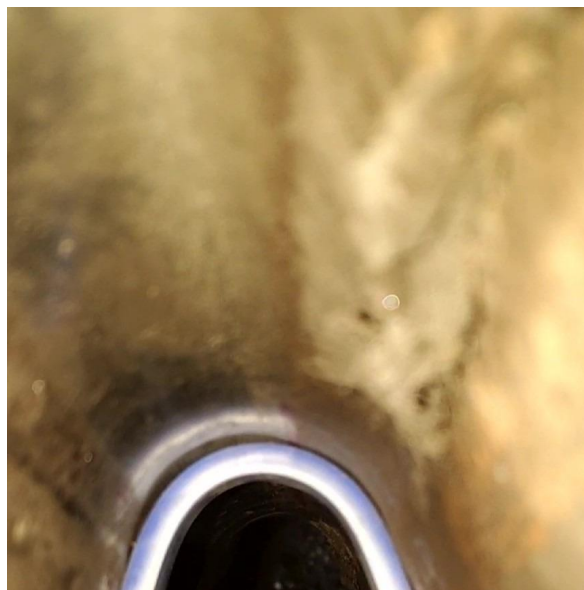


PHOTO-DGN-002
Profile-gauge view showing the worn Deflector-Guide Sheave
North groove.
Related defect: DEF-003

Figure 31 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-001 / PHOTO-DGN-002

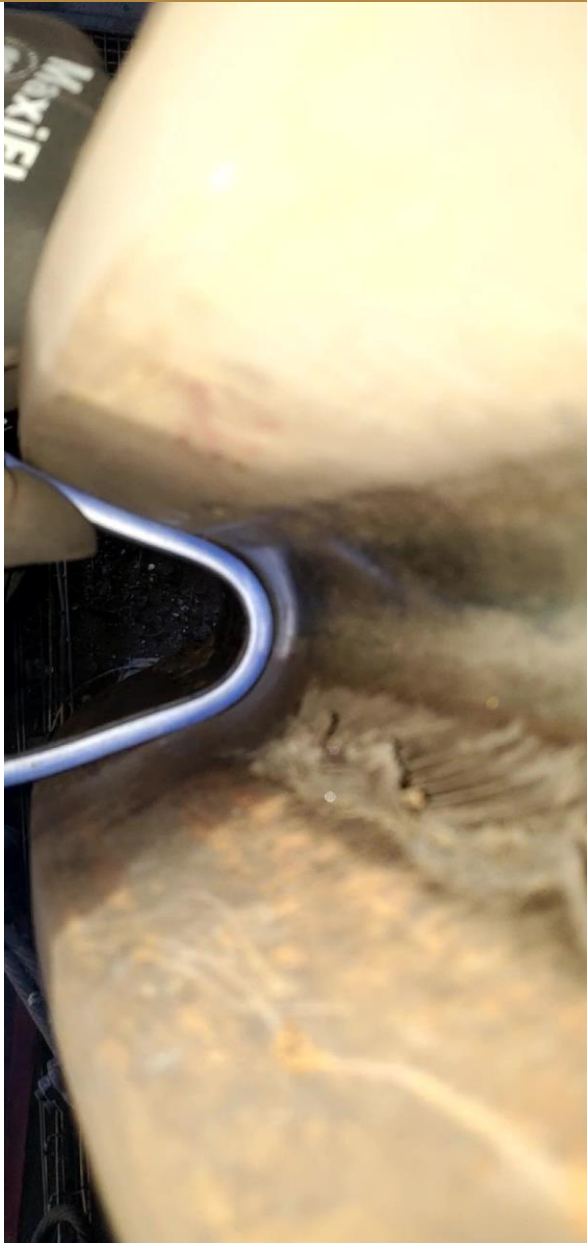


PHOTO-DGN-003
Additional profile-gauge view of the worn Deflector-Guide Sheave North groove.
Related defect: DEF-003



PHOTO-DGN-004
Groove surface detail showing asymmetric polishing/wear on Deflector-Guide Sheave North.
Related defect: DEF-003

Figure 32 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-003 / PHOTO-DGN-004

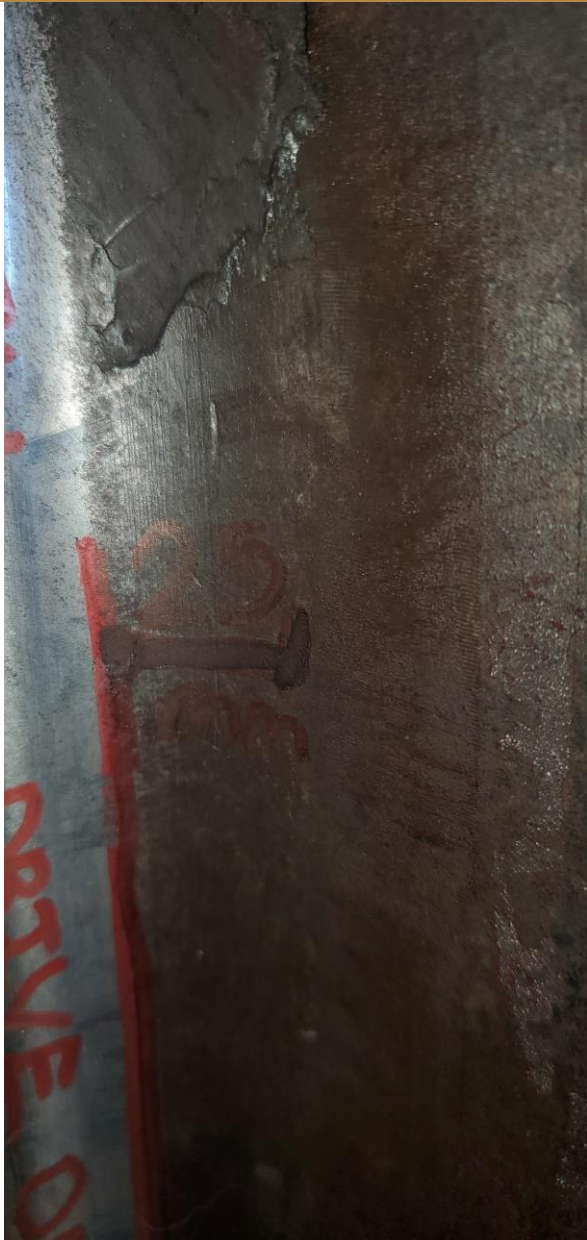


PHOTO-DGN-005
Marked groove view illustrating lateral offset of the worn contact
profile - North.
Related defect: DEF-003



PHOTO-DGN-006
Additional marked view illustrating lateral offset of the worn contact
profile - North.
Related defect: DEF-003

Figure 33 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-005 / PHOTO-DGN-006



PHOTO-DGN-007

Contextual side view of the offset groove profile - North.

Related defect: DEF-003

Figure 34 - Deflector-Guide Sheave North photographic evidence - PHOTO-DGN-007

Deflector-Guide Sheave South



PHOTO-DGS-001
Profile-gauge view showing the worn Deflector-Guide Sheave South groove.
Related defect: DEF-004



PHOTO-DGS-002
Additional profile-gauge view of the Deflector-Guide Sheave South groove.
Related defect: DEF-004

Figure 35 - Deflector-Guide Sheave South photographic evidence - PHOTO-DGS-001 / PHOTO-DGS-002



PHOTO-DGS-003

Groove surface detail showing the wear profile on Deflector-Guide Sheave South.

Related defect: Information / context only

PHOTO-DGS-004

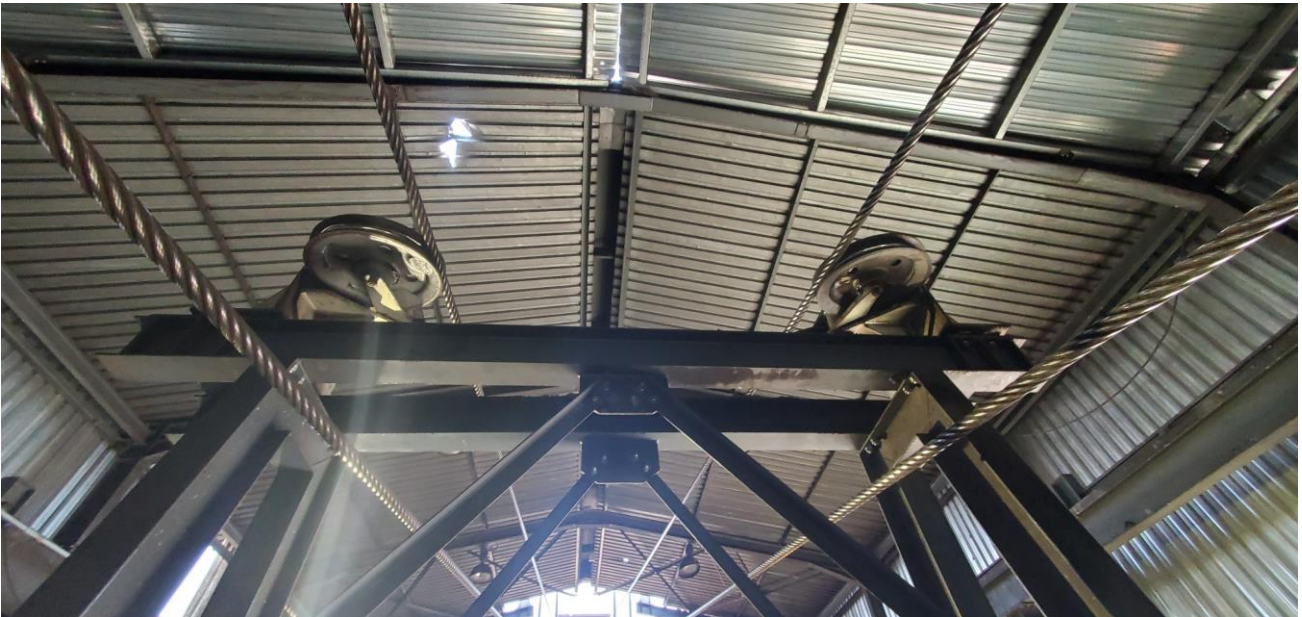
Marked view illustrating the offset/asymmetric groove contact profile - South.

Related defect: DEF-004

Figure 36 - Deflector-Guide Sheave South photographic evidence - PHOTO-DGS-003 / PHOTO-DGS-004

Cable Belt Upper Incoming Guide Sheaves

The following photographs are included as contextual reference views of the Cable Belt Upper Incoming Guide Sheaves. They identify the physical North and South guide-sheave assemblies that control the rope approach toward the Drive Wheel sheaves and may require positional review if a non-zero rope-entry fleet angle is confirmed by formal survey. No alignment value or corrective repositioning requirement is inferred from these photographs alone.



CBIGS-001

Contextual underside view showing the North and South Cable Belt Upper Incoming Guide Sheaves and the cable-belt rope paths approaching the main drive arrangement.

Related defect: Information / system context only

Figure 37 - Cable Belt Upper Incoming Guide Sheaves contextual arrangement - CBIGS-001



CBIGS-002

Contextual view of the Cable Belt Upper Incoming Guide Sheave North and the associated incoming rope path.

Related defect: Information / system context only



CBIGS-003

Contextual view of the Cable Belt Upper Incoming Guide Sheave South and the associated incoming rope path.

Related defect: Information / system context only

Figure 38 - Cable Belt Upper Incoming Guide Sheaves North / South - CBIGS-002 / CBIGS-003

Appendix B - Inspection Checklist

Table 52 - Cable belt sheave inspection checklist

Asset	Inspection check	Result	Observation / limitation	Evidence reference	Related defect ID
Drive Wheel North	Asset/location identified	Pass - photographic evidence	Location and asset role supported by photo set / CBDS-REF001.	Relevant PHOTO series / CBDS-REF001	Information / DEF-001
Drive Wheel North	Rope groove / liner condition examined	Fail / corrective action	Asymmetric wear or groove/liner damage recorded.	Relevant PHOTO series	DEF-001
Drive Wheel North	Rope tracking / alignment indicators examined	Fail / corrective action	Evidence supports off-center rope tracking; formal alignment measurement outstanding.	Relevant PHOTO series / CBDS-REF002/003	DEF-001
Drive Wheel North	Hub, shaft and bearing condition externally assessed	Partial	No sufficient bearing-condition data for final conclusion; follow-up required.	Context photos where available	DEF-001
Drive Wheel South	Asset/location identified	Pass - photographic evidence	Location and asset role supported by photo set / CBDS-REF001.	Relevant PHOTO series / CBDS-REF001	Information / DEF-002
Drive Wheel South	Rope groove / liner condition examined	Fail / corrective action	Asymmetric wear or groove/liner damage recorded.	Relevant PHOTO series	DEF-002
Drive Wheel South	Rope tracking / alignment indicators examined	Fail / corrective action	Evidence supports off-center rope tracking; formal alignment measurement outstanding.	Relevant PHOTO series / CBDS-REF002/003	DEF-002
Drive Wheel South	Hub, shaft and bearing condition externally assessed	Partial	No sufficient bearing-condition data for final conclusion; follow-up required.	Context photos where available	DEF-002
Deflector-Guide Sheave North	Asset/location identified	Pass - photographic evidence	Location and asset role supported by photo set / CBDS-REF001.	Relevant PHOTO series / CBDS-REF001	Information / DEF-003
Deflector-Guide Sheave North	Rope groove / liner condition examined	Fail / corrective action	Asymmetric wear/off-center profile recorded. Field notes record 65 mm groove depth and approximately 3 mm local wear; remaining service margin requires verified OEM/client criteria.	PHOTO-DGN series; inspection field notes	DEF-003
Deflector-Guide Sheave North	Rope tracking / alignment indicators examined	Fail / corrective action	Evidence supports off-center rope tracking; formal alignment measurement outstanding.	Relevant PHOTO series / CBDS-REF002/003	DEF-003

Asset	Inspection check	Result	Observation / limitation	Evidence reference	Related defect ID
Deflector-Guide Sheave North	Hub, shaft and bearing condition externally assessed	Partial	No sufficient bearing-condition data for final conclusion; follow-up required.	Context photos where available	DEF-003
Deflector-Guide Sheave South	Asset/location identified	Pass - photographic evidence	Location and asset role supported by photo set / CBDS-REF001.	Relevant PHOTO series / CBDS-REF001	Information / DEF-004
Deflector-Guide Sheave South	Rope groove / liner condition examined	Fail / corrective action	Asymmetric wear/off-center profile recorded. Field notes record 66 mm groove depth and approximately 2 mm local wear; remaining service margin requires verified OEM/client criteria.	PHOTO-DGS series; inspection field notes	DEF-004
Deflector-Guide Sheave South	Rope tracking / alignment indicators examined	Fail / corrective action	Evidence supports off-center rope tracking; formal alignment measurement outstanding.	Relevant PHOTO series / CBDS-REF002/003	DEF-004
Deflector-Guide Sheave South	Hub, shaft and bearing condition externally assessed	Partial	No sufficient bearing-condition data for final conclusion; follow-up required.	Context photos where available	DEF-004

Appendix C - Measurements and Dimensional Records

MEASUREMENT STATUS

Formal rope-path alignment/fleet-angle measurements and Drive Wheel remaining liner-thickness measurements remain outstanding. Inspection field notes do, however, record selected as-found Deflector-Guide groove measurements: North groove depth 65 mm with approximately 3 mm local wear, and South groove depth 66 mm with approximately 2 mm local wear. The North 65 mm value is additionally traceable to the handwritten marking visible in PHOTO-DGN-004 (Figure 21), and the approximately 3 mm North wear location is indicated in PHOTO-DGN-002 (Figure 19). No photographs of the measuring operations were retained. The Bosch PLR 50, equipment ID 009741746, is recorded for the groove-depth measurements; however, the measurement method/instrument used for the approximate local-wear values and the exact indexed measurement positions were not recorded. MEAS-001 through MEAS-008 are retained as the structured follow-up measurement schedule generated by this report; MEAS-009 through MEAS-012 record the available as-found field readings. The as-found readings are condition evidence and are not stand-alone acceptance/discard criteria without verified OEM/client groove geometry.

Table 53 - Measurements and dimensional records

Measurement ID	Asset	Location / feature	Instrument / equipment ID	Measured value	Unit	Reference / criterion	Deviation	Comment
MEAS-001	North incoming rope path	Cable Belt Upper Incoming Guide Sheave North to Drive Wheel North entry fleet angle	Laser/optical alignment equipment - ID to be recorded	Not recorded	deg	Verified design/OEM/client fleet-angle tolerance; target 0 deg or as close as practicable	Not calculable	Quantifies rope-entry geometry represented by CBDS-REF009/010 and CBIGS-001/002.
MEAS-002	South incoming rope path	Cable Belt Upper Incoming Guide Sheave South to Drive Wheel South entry fleet angle	Laser/optical alignment equipment - ID to be recorded	Not recorded	deg	Verified design/OEM/client fleet-angle tolerance; target 0 deg or as close as practicable	Not calculable	Quantifies rope-entry geometry represented by CBDS-REF009/010 and CBIGS-001/003.
MEAS-003	North sheave pair	Drive Wheel North exit to Deflector-Guide Sheave North entry / centerline relationship	Laser/optical alignment equipment - ID to be recorded	Not recorded	mm / deg	Verified design centerline / client-OEM tolerance	Not calculable	Required to quantify DEF-001 / DEF-003; refer CBDS-REF002/003 /004.
MEAS-004	South sheave pair	Drive Wheel South exit to Deflector-Guide Sheave South entry / centerline relationship	Laser/optical alignment equipment - ID to be recorded	Not recorded	mm / deg	Verified design centerline / client-OEM tolerance	Not calculable	Required to quantify DEF-002 / DEF-004; refer CBDS-REF002/003 /004.
MEAS-005	Deflector-Guide Sheave North downstream path	Deflector-Guide Sheave North exit fleet angle	Laser/optical alignment equipment - ID to be recorded	Not recorded	deg	Verified design/OEM/client fleet-angle tolerance; target 0 deg or as close as practicable	Not calculable	Required to verify the possible exit-side mechanism; refer CBDS-REF011/012 /013.

Measurement ID	Asset	Location / feature	Instrument / equipment ID	Measured value	Unit	Reference / criterion	Deviation	Comment
MEAS-006	Deflector-Guide Sheave South downstream path	Deflector-Guide Sheave South exit fleet angle	Laser/optical alignment equipment - ID to be recorded	Not recorded	deg	Verified design/OEM/client fleet-angle tolerance; target 0 deg or as close as practicable	Not calculable	Required to verify the possible exit-side mechanism; refer CBDS-REF011/012 /013.
MEAS-007	Drive Wheels North and South	Groove/liner profile and remaining material / replacement baseline	Approved profile/dimensional gauge - ID to be recorded	Not recorded	mm	Verified OEM/client liner geometry and installation limits	Not calculable	Record pre-removal condition where practicable and post-renewal baseline.
MEAS-008	Deflector-Guide Sheaves North and South	Existing groove profile and replacement-sheave baseline geometry	Approved groove profile gauge - ID to be recorded	Not recorded	mm	Verified OEM/client groove geometry	Not calculable	Existing grooves are recommended for replacement; record incoming replacement geometry and post-installation baseline.
MEAS-009	Deflector-Guide Sheave North	Rope groove depth - as-found field reading	Bosch PLR 50 / 009741746	65	mm	Verified nominal/new and discard groove-depth criterion not supplied	Not calculable	Recorded in inspection field notes; handwritten 65 mm value visible on sheave in PHOTO-DGN-004 / Figure 21. No measurement-operation photograph retained.
MEAS-010	Deflector-Guide Sheave North	Local groove wear - indicated inspection location	Instrument / equipment ID not recorded	~3	mm	Verified OEM/client local groove-wear/profile criterion not supplied	Not calculable	Approximate field-note reading; wear location indicated in PHOTO-DGN-002 / Figure 19. Repeat at indexed position with recorded equipment.

Measurement ID	Asset	Location / feature	Instrument / equipment ID	Measured value	Unit	Reference / criterion	Deviation	Comment
MEAS-011	Deflector-Guide Sheave South	Rope groove depth - as-found field reading	Bosch PLR 50 / 009741746	66	mm	Verified nominal/new and discard groove-depth criterion not supplied	Not calculable	Recorded in inspection field notes. No measurement-operation photograph retained; repeat at indexed position with recorded equipment.
MEAS-012	Deflector-Guide Sheave South	Local groove wear - inspection field reading	Instrument / equipment ID not recorded	~2	mm	Verified OEM/client local groove-wear/profile criterion not supplied	Not calculable	Approximate field-note reading. No measurement-operation photograph retained; repeat at indexed position with recorded equipment.

Appendix D - NDT Results

No NDT results were supplied for inclusion in this report. Where NDT is subsequently performed as part of an approved follow-up scope, the method, scope, operator certification, applicable procedure and result evidence should be retained in the corrective-action and close-out records.

Table 54 - NDT results register

NDT ID	Asset	Location	Method	Procedure / technique	Operator / certification	Result	Report reference	Comment
N/A	All four sheaves	Not applicable	Not supplied	N/A	N/A	No result reported	N/A	Retain as not used unless final scope confirms NDT.

Appendix E - Drawings, Schematics and Reference Information

Table 55 - Drawings and schematics register

Reference ID	Drawing / document number	Title	Revision	Source	Status	Comment
CBDS-REF001	CBDS-REF001	Cable belt drive sheave arrangement and inspection locations.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF002	CBDS-REF002	Contextual misalignment mechanism with indicative wear locations.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF003	CBDS-REF003	Contextual misalignment between the Drive Wheel and Deflector-Guide Sheave.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF004	CBDS-REF004	Desired alignment with the rope path coincident with the sheave centerlines.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF005	CBDS-REF005	Illustrative left-side groove wear produced by angular rope tracking.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF007	CBDS-REF007	Illustrative bearing-load mechanism associated with a misaligned sheave.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF008	CBDS-REF008	Reference groove profile and nominal symmetrical wear pattern illustration.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF009	CBDS-REF009	Desired Drive Wheel Rope Entry Path.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF010	CBDS-REF010	Contextual Drive Wheel Rope Entry Misalignment.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF011	CBDS-REF011	Desired fleet angle of the rope exiting the Deflector-Guide Sheaves.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.

Reference ID	Drawing / document number	Title	Revision	Source	Status	Comment
CBDS-REF012	CBDS-REF012	Example of a possible non-zero fleet angle of the rope exiting the Deflector-Guide Sheaves.	1.0	ISS / supplied reference package	Reference only	Schematic / not to scale; used in Section 14.
CBDS-REF013	CBDS-REF013	Proposed introduction of lower overhead guide sheaves to manage the exit fleet angle from the Deflector-Guide Sheaves.	1.0	ISS / supplied reference package	Reference only	Conceptual schematic / not to scale; used in Section 14 and later recommendations.

Additional supplied illustrations CBDS-REF006 and Sheave Groove Cross Section were retained in the source package but were not specifically instructed for insertion in this report section. CBDS-REF009 to CBDS-REF013 have been inserted in Section 14 to support the later system-level engineering assessment, recommendations and conclusion discussion regarding rope-entry alignment, Deflector-Guide Sheave exit fleet angle and the conceptual use of lower overhead guide sheaves to control the exit rope path.

Appendix F - Supporting Documentation

Table 56 - Supporting documentation register

Document ID	Title / description	Document number	Revision / date	Source	Report relevance	Status
DOC-001	Cable Belt Drive Sheaves Report Summary Notes and Compile Instructions	Compilation notes	Supplied 07/08/2026	ISS / project working file	Primary source for scope, known asset data, image selection and initial inspection findings	Included
DOC-002	Reference Drawings image set	CBDS-REF image set	Rev 1.0	ISS	Supports explanatory arrangement, alignment and wear mechanism illustrations	Included
DOC-003	Drive Wheel North photo set	DWN image set	18/07/2026	ISS	Supports DEF-001	Included
DOC-004	Drive Wheel South photo set	DWS image set	18/07/2026	ISS	Supports DEF-002	Included
DOC-005	Deflector-Guide Sheave North photo set	DGSN image set	18/07/2026	ISS	Supports DEF-003	Included
DOC-006	Deflector-Guide Sheave South photo set	DGSS image set	18/07/2026	ISS	Supports DEF-004	Included
DOC-007	Alignment survey / measurement sheet	Not supplied	Required current record	South32 / contractor	Primary post-issue action to quantify rope-path geometry and support corrective-action close-out.	Outstanding
DOC-008	OEM groove/liner acceptance criteria	Not supplied	Current revision required	South32 / OEM	Required to verify applicable groove/liner acceptance criteria and support component-renewal and corrective-action close-out	Outstanding
DOC-009	Cable Belt Upper Incoming Guide Sheave contextual photo set	CBIGS image set	18/07/2026	ISS	Provides visual identification/context for the upper incoming guide sheaves used in the system-level rope-entry fleet-angle review.	Included
DOC-010	Inspection field notes - Deflector-Guide groove measurements	ISS inspector field notes	18/07/2026	ISS / inspector field record	Source record for North 65 mm groove depth / approximately 3 mm local wear and South 66 mm groove depth / approximately 2 mm local wear.	Referenced - field notes not reproduced in this appendix